

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Number Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

NUMBER TOOLKIT · CH1 · L1

ENG. ESLAM AHMED



The Map

This lesson collapses Number Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Prime factor structure

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Standard form conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Fractions and decimals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Rounding and bounds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the LCM

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 106 website questions, 106 checked solutions, 3 local PDFs read.

01 Prime factor structure

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

2 Find the lowest common multiple (LCM) of 28 and 105

YOUR SOLUTION

02 Standard form conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5.6 \times 10^7 \div 10^2 \times 8 \times 10^4 \times 3 \times 10^{-3}$. Give your answer in standard form. (2)

YOUR SOLUTION

03 Fractions and decimals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

2 Show that $315 \text{ times } 258 = 825$

YOUR SOLUTION

04 Rounding and bounds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

06 Find the LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Prime factor structure
"work out", "show", "hence", a familiar exam phrase	Standard form conversion
"work out", "show", "hence", a familiar exam phrase	Fractions and decimals
"work out", "show", "hence", a familiar exam phrase	Rounding and bounds
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "find", a missing value	Find the LCM

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Set Notation & Venn Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SET NOTATION & VENN DIAGRAMS · CH1 · L2

ENG. ESLAM AHMED



The Map

This lesson collapses Set Notation & Venn Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Fill the intersection first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use union and intersection

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use complements

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Read probability from the diagram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 23 website questions, 23 checked solutions, 3 local PDFs read.

01 Fill the intersection first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q4

Calculate value

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

means all the letters that

$$B \cup G = \{b, l, u, e, g, r, y\}$$

Finish: (a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

4. $B = \{b, l, u, e\}$ $G = \{g, r, e, y\}$ $W = \{w, h, i, t, e\}$ (a) List all the members of the set (i) $B \cup G$ (ii) $W \cap G$ (2) Serena writes down the statement $B \cap G \cap W = \text{empty set}$ (b) Is Serena's statement correct? You must give a reason for your answer. (1)

YOUR SOLUTION

02 Use union and intersection

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$G \cap S \text{ only} = 11 - 6 = 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Specimen 4WM1H Q14

Place the centre value

This is a reasoning step: write one clear sentence, then continue the calculation.

Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

Finish: Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 100 farmers are asked if they have goats, sheep or chickens. Complete the Venn diagram, find $n(G)$, $n((G \cup S)')$, $n(G' \cap C)$, and the probability a farmer with chickens also has goats.

YOUR SOLUTION

03 Use complements

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q4

Calculate value

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

The odd numbers are

$$A = \{21, 23, 25, 27, 29\}$$

Finish: (i) $\{21, 27\}$, (ii) $\{21, 23, 24, 25, 27, 29\}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 E = {20, 21, 22, 23, 24, 25, 26, 27, 28, 29} A = {odd numbers} B = {multiples of 3} List the members of the set (i) A intersection B (1) (ii) A union B (1)

YOUR SOLUTION

04 Read probability from the diagram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$G \cap S \text{ only} = 11 - 6 = 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Specimen 4WM1H Q14

Place the centre value

This is a reasoning step: write one clear sentence, then continue the calculation.

Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

Finish: Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

14 100 farmers are asked if they have goats, sheep or chickens. Complete the Venn diagram, find $n(G)$, $n((G \cup S)')$, $n(G' \cap C)$, and the probability a farmer with chickens also has goats.

YOUR SOLUTION

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q4

Calculate value

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

means all the letters that

$$B \cup G = \{b, l, u, e, g, r, y\}$$

Finish: (a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

4 B = {b, l, u, e} G = {g, r, e, y} W = {w, h, i, t, e} (a) List all the members of the set (i) B union G (ii) W intersection G ' (2) Serena writes down the statement B intersection G intersection W = empty set (b) Is Serena's statement correct? You must give a reason for your answer. (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Fill the intersection first
"work out", "show", "hence", a familiar exam phrase	Use union and intersection
"work out", "show", "hence", a familiar exam phrase	Use complements
"work out", "show", "hence", a familiar exam phrase	Read probability from the diagram
"work out", "show", "hence", a familiar exam phrase	Calculate statistic

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Prime Factors, HCF & LCM

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PRIME FACTORS, HCF & LCM · CH1 · L3



ENG. ESLAM AHMED

The Map

This lesson collapses Prime Factors, HCF & LCM into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Write prime factor trees

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find HCF

TRIGGER *"work out", "find", a missing value*

03 Find LCM

TRIGGER *"work out", "find", a missing value*

04 Use HCF/LCM in context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the LCM

TRIGGER *"work out", "find", a missing value*

06 The HCF takes the smaller

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 25 website questions, 25 checked solutions, 3 local PDFs read.

01 Write prime factor trees

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5^{17} \times 5^2 = 5^{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q1

Use the index law

$$5^{17} \times 5^2 = 5^{19}$$

Find the LCM

$$800 = 8 \times 100$$

Finish: (a) 5^{19} , (b) $2^5 \times 5^2$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

- 1 (a) Write 517 times 52 as a single power of 5 (1) (b) Write 800 as a product of its prime factors. Show your working clearly. (2)

YOUR SOLUTION

02 Find HCF

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5^{17} \times 5^2 = 5^{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q1

Use the index law

$$5^{17} \times 5^2 = 5^{19}$$

Find the LCM

$$800 = 8 \times 100$$

Finish: (a) 5^{19} , (b) $2^5 \times 5^2$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

- 1 (a) Write 517 times 52 as a single power of 5 (1) (b) Write 800 as a product of its prime factors. Show your working clearly. (2)

YOUR SOLUTION

03 Find LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^8 \times 3^5 \times 11^4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q9

Find the LCM

$$A = 2^8 \times 3^5 \times 11^4$$

Find the LCM

$$\text{HCF} = 2^6 \times 3 \times 11^4$$

Finish: (a) $2^6 \times 3 \times 11^4$, (b) $2^9 \times 3^5 \times 11^8$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

9 A = 2 8 times 3 5 times 11 4 B = 2 6 times 3 times 11 8 (a) Find the highest common factor (HCF) of A and B. (2) (b) Find the lowest common multiple (LCM) of 2A and 3B. Give the LCM as a product of powers of its prime factors. (2)

YOUR SOLUTION

04 Use HCF/LCM in context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q9

Find the LCM

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

Find the LCM

$$\text{LCM} = 2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

Finish: $2^4 \times 3^2 \times 5^4 \times 11 \times 13$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

9 A = 23 times 32 times 52 times 11 B = 24 times 3 times 54 times 13 Find the lowest common multiple (LCM) of A and B. Give your answer as a product of powers of prime numbers.

YOUR SOLUTION

05 Find the LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 The HCF takes the smaller

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^5 \times 5 \times 7^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1HR Q11

Find the LCM

$$A = 2^5 \times 5 \times 7^2$$

Find the LCM

$$5A = 2^5 \times 5^2 \times 7^2$$

Finish: (a) $2^4 \times 5^2 \times 7^2$, (b) $2^{16} \times 5^8 \times 7^{12}$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

11 A B 2 5 7 2 5 7 5 2 3 3 4 (a) Write down the highest common factor (HCF) of 5A and 2B Give your answer as a product of prime factors. (2) A B 2 5 7 2 5 7 5 2 3 3 4 (b) Work out the value of AB 2 Give your answer as a product of prime factors. (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Write prime factor trees
"work out", "find", a missing value	Find HCF
"work out", "find", a missing value	Find LCM
"work out", "show", "hence", a familiar exam phrase	Use HCF/LCM in context
"work out", "find", a missing value	Find the LCM
"work out", "show", "hence", a familiar exam phrase	The HCF takes the smaller

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Powers, Roots & Standard Form

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

POWERS, ROOTS & STANDARD FORM · CH1 · L4



ENG. ESLAM AHMED

The Map

This lesson collapses Powers, Roots & Standard Form into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Handle negative and fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Calculate then return to standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Difference

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Write this in standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 30 website questions, 30 checked solutions, 3 local PDFs read.

01 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

02 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM2H Q13

Use index laws

This is a reasoning step: write one clear sentence, then continue the calculation.

Solve inequality

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 7.29 times 10^{242}

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Handle negative and fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out 5610710281043
 $3 \dots$ times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

04 Calculate then return to standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.357 \times 10^5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q8

The greatest coastline is the

$$1.357 \times 10^5$$

Convert standard form

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) Pacific Ocean, (b) 9.393×10^4 km

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

8 The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean Length of coastline (km) Arctic 4.539×10^4 Atlantic 1.119×10^5 Pacific 1.357×10^5 Indian 6.653×10^4 Southern 1.797×10^4 (a) Which ocean has the greatest length of coastline? (1) (b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline. Give your answer in standard form. km (2)

YOUR SOLUTION

05 Difference

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.06 \times 10^8 = 106000000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q10

Atlantic surface area

$$1.06 \times 10^8 = 106000000$$

Indian surface area

$$6.86 \times 10^7 = 68600000$$

Finish: (a) 3.74×10^7 , (b) 11

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean. Give your answer in standard form. square kilometres

(2) The surface area of the Pacific Ocean is k times the surface area of the ...

YOUR SOLUTION

06 Write this in standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 4.2 \times 10^{-24}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q8

Convert standard form

$$a = 4.2 \times 10^{-24}$$

Convert standard form

$$ab = (4.2 \times 3) \times 10^{-24+145}$$

Finish: 1.26×10^{122}

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

8 a = 4.2 times 10^{-24} b = 3 times 10^{145} Work out the value of a times b Give your answer in standard form.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Handle negative and fractional powers
"work out", "show", "hence", a familiar exam phrase	Calculate then return to standard form
"work out", "show", "hence", a familiar exam phrase	Difference
"work out", "show", "hence", a familiar exam phrase	Write this in standard form

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Fractions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FRACTIONS · CH1 · L5

ENG. ESLAM AHMED



The Map

This lesson collapses Fractions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use a common denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Multiply and divide fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Simplify mixed numbers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Convert the final form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Convert to improper fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Use a common denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4\frac{2}{3} = \frac{14}{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q3

Convert to improper fractions

$$4\frac{2}{3} = \frac{14}{3}$$

Use a common denominator

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

Finish: $8\frac{7}{15}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Multiply and divide fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Simplify mixed numbers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{2}{3} = \frac{11}{3}, \quad 2\frac{4}{7} = \frac{18}{7}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1H Q1

Simplify fraction

$$3\frac{2}{3} = \frac{11}{3}, \quad 2\frac{4}{7} = \frac{18}{7}$$

Simplify fraction

$$\frac{11}{3} \times \frac{18}{7} = \frac{198}{21} = \frac{66}{7}$$

Finish: $9\frac{3}{7}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

1 Without using a calculator, work out $3\frac{2}{3} \times 2\frac{4}{7}$ times. Show each stage of your working. Give your answer as a mixed number in its simplest form.

YOUR SOLUTION

04 Convert the final form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6\frac{3}{4} = \frac{27}{4}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q2

Convert to improper fractions

$$6\frac{3}{4} = \frac{27}{4}$$

Divide by multiplying by the

$$\frac{27}{4} \div \frac{18}{7} = \frac{27}{4} \times \frac{7}{18}$$

Finish: $2\frac{5}{8}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Convert to improper fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4\frac{2}{3} = \frac{14}{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q3

Convert to improper fractions

$$4\frac{2}{3} = \frac{14}{3}$$

Use a common denominator

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

Finish: $8\frac{7}{15}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use a common denominator
"work out", "show", "hence", a familiar exam phrase	Multiply and divide fractions
"work out", "show", "hence", a familiar exam phrase	Simplify mixed numbers
"work out", "show", "hence", a familiar exam phrase	Convert the final form
"work out", "show", "hence", a familiar exam phrase	Simplify fraction
"work out", "show", "hence", a familiar exam phrase	Convert to improper fractions

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Percentages

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PERCENTAGES · CH1 · L6



ENG. ESLAM AHMED

The Map

This lesson collapses Percentages into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Percentage profit

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 40 website questions, 40 checked solutions, 3 local PDFs read.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$100\% - 15\% = 85\%$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q7

The hat is reduced by

$$100\% - 15\% = 85\%$$

of the normal price

$$0.85 \times \text{normal price} = 20.40$$

Finish: 24 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$100\% - 15\% = 85\%$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q7

The hat is reduced by

$$100\% - 15\% = 85\%$$

of the normal price

$$0.85 \times \text{normal price} = 20.40$$

Finish: 24 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times(1 + \frac{r}{100})$, decrease means $\times(1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

7 In a sale, the normal price of a hat is reduced by 15% The sale price of the hat is 20.40 euros. Work out the normal price of the hat. euros

YOUR SOLUTION

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$14 \times 7 \times 5 = 490$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q6

Pierre's old weekly pay

This is a reasoning step: write one clear sentence, then continue the calculation.

His new weekly pay is

$$14 \times 7 \times 5 = 490$$

Finish: Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week. In his new job, Pierre is paid 14 euros per hour. In his new job, Pierre works for 7 hours each day for a 5-day week. (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week. % (4) Marie changes her job. Her salary decreases by 6% Her new salary is 23 030 euros. (b) Work out Marie's salary before she changes her job. euros (3)

YOUR SOLUTION

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$14 \times 7 \times 5 = 490$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q6

Pierre's old weekly pay

This is a reasoning step: write one clear sentence, then continue the calculation.

His new weekly pay is

$$14 \times 7 \times 5 = 490$$

Finish: Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week. In his new job, Pierre is paid 14 euros per hour. In his new job, Pierre works for 7 hours each day for a 5-day week. (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week. % (4) Marie changes her job. Her salary decreases by 6% Her new salary is 23 030 euros. (b) Work out Marie's salary before she changes her job. euros (3)

YOUR SOLUTION

05 Calculate percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$545 - 500 = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q8

Convert standard form

$$545 - 500 = 45$$

Convert standard form

$$592 - 545 = 47$$

Finish: Brigitte is not correct; the discount is 168 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times(1 + \frac{r}{100})$, decrease means $\times(1 - \frac{r}{100})$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says, "The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017"

(a) Is Brigitte correct? You must show how you get your answer. (4)

Henri wants to insure his car. He gets a discount of 15% off the normal price. Henri pays 952 euros for his ...

YOUR SOLUTION

06 Percentage profit

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$520 - 465 = 55$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q3

Profit

$$520 - 465 = 55$$

Percentage profit

$$\frac{55}{465} \times 100 = 11.827 \dots$$

Finish: (a) 11.8%, (b) \$484

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

3 Gladys buys a table for \$465 to sell in her shop. She sells the table for \$520 (a) Work out the percentage profit that Gladys makes from the sale of the table. Give your answer correct to 3 significant figures. % (3) Gladys has a sale in her shop. She decreases all the normal prices by 12% The normal price of an armchair was \$550 (b) Work out the sale price of the armchair. \$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Reverse a percentage
"work out", "show", "hence", a familiar exam phrase	Use compound change
"work out", "show", "hence", a familiar exam phrase	Compare percentage change
"work out", "show", "hence", a familiar exam phrase	Calculate percentage
"work out", "show", "hence", a familiar exam phrase	Percentage profit

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Compound Interest & Depreciation

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMPOUND INTEREST & DEPRECIATION · CH1 · L7



ENG. ESLAM AHMED

The Map

This lesson collapses Compound Interest & Depreciation into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 The multiplier for compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 32 website questions, 32 checked solutions, 3 local PDFs read.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $0.85V$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q14

Let the original house value

This is a reasoning step: write one clear sentence, then continue the calculation.

Use compound interest

| $0.85V$

Finish: 10%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $0.85V$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q14

Let the original house value

This is a reasoning step: write one clear sentence, then continue the calculation.

Use compound interest

| $0.85V$

Finish: 10%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 Simon bought a house at the beginning of 2018. The value of Simon's house had decreased by 15% by the end of 2018. The house increased in value during both 2019 and 2020. The percentage increases in the value of the house during 2019 and 2020 were the same. The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018. Calculate the percentage increase in the value of the house during 2019 %

YOUR SOLUTION

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6000(1.015)^2 = 6181.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q11

Use compound interest

$$6000(1.015)^2 = 6181.35$$

Let the third year interest

$$6181.35 \left(1 + \frac{r}{100}\right) = 6311.16$$

Finish: 2.1%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

11 Max invests \$6000 in a savings account for 3 years. The account pays compound interest at a rate of 1.5% per year for the first 2 years. The compound interest rate changes for the third year. At the end of 3 years, there is a total of \$6311.16 in the account. Work out the compound interest rate for the third year. Give your answer correct to 1 decimal place. %

YOUR SOLUTION

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q6

Use compound interest

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

Use compound interest

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) No, it stayed in band B. (b) 27.75%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

6 Asha bought an apartment. The table gives information about the value of apartments, in euros, and the annual service charge band. Value (x euros) Service charge band
 $x \geq 700\,000$ A $600\,000 < x < 700\,000$ B $500\,000 < x < 600\,000$ C $400\,000 < x < 500\,000$ D $0 < x < 400\,000$ E
 In 2021, the value of Asha's apartment was 634 400 euros. The value of Asha's apartment had increased by 4% from its value in 2020 (a) Has the annual service charge band changed for Asha's apartment? Show your working clearly.
 (3) Pam bought a ...

YOUR SOLUTION

05 Use compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6000(1.015)^2 = 6181.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q11

Use compound interest

$$6000(1.015)^2 = 6181.35$$

Let the third year interest

$$6181.35 \left(1 + \frac{r}{100}\right) = 6311.16$$

Finish: 2.1%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

11 Max invests \$6000 in a savings account for 3 years. The account pays compound interest at a rate of 1.5% per year for the first 2 years. The compound interest rate changes for the third year. At the end of 3 years, there is a total of \$6311.16 in the account. Work out the compound interest rate for the third year. Give your answer correct to 1 decimal place. %

YOUR SOLUTION

06 The multiplier for compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1.025

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q5

The multiplier for compound interest

1.025

Use compound interest

$$7200(1.025)^3 = 7753.6125$$

Finish: 7754 dollars

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 Shane invests 7200 dollars for 3 years in a savings account. He gets 2.5% per year compound interest. How much money will Shane have in his savings account at the end of 3 years? Give your answer to the nearest dollar. dollars

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Reverse a percentage
"work out", "show", "hence", a familiar exam phrase	Use compound change
"work out", "show", "hence", a familiar exam phrase	Compare percentage change
"work out", "show", "hence", a familiar exam phrase	Use compound interest
"work out", "show", "hence", a familiar exam phrase	The multiplier for compound interest

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Fractions, Decimals & Percentages

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FRACTIONS, DECIMALS & PERCENTAGES · CH1 · L8



ENG. ESLAM AHMED

The Map

This lesson collapses Fractions, Decimals & Percentages into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Convert fraction to decimal

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Convert decimal to percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare all values in one form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 1 website questions, 1 checked solutions, 3 local PDFs read.

01 Convert fraction to decimal

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4\dot{x}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4\dot{x}$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

18 $0.4x$ is a recurring decimal. x is a whole number such that $1 < x < 9$. Find, in terms of x , the recurring decimal $0.4x$ as a fraction. Give your fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

02 Convert decimal to percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Compare all values in one form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

18 $0.4x$ is a recurring decimal. x is a whole number such that $1 < x < 9$. Find, in terms of x , the recurring decimal $0.4x$ as a fraction. Give your fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

06 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Convert fraction to decimal
"work out", "show", "hence", a familiar exam phrase	Convert decimal to percentage
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Compare all values in one form
"work out", "show", "hence", a familiar exam phrase	Simplify fraction
"work out", "show", "hence", a familiar exam phrase	Subtract

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Rounding, Estimation & Bounds

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ROUNDING, ESTIMATION & BOUNDS · CH1 · L9



ENG. ESLAM AHMED

The Map

This lesson collapses Rounding, Estimation & Bounds into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Round to the requested accuracy

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find lower and upper bounds

TRIGGER *"work out", "find", a missing value*

03 Choose bounds for largest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Choose bounds for smallest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find upper bound

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 37 website questions, 37 checked solutions, 3 local PDFs read.

01 Round to the requested accuracy

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$X = \text{area of rectangle} - \text{area of square}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2H Q22

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

Bounds

$$11.25 \leq AB < 11.75$$

Finish: $X = 90 \text{ cm}^2$ to a suitable degree of accuracy

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

22 The diagram shows a square inside rectangle ABCD. A B D C Diagram NOT accurately drawn. The total area of the region shown shaded in the diagram is $X \text{ cm}^2$. $AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm . $BC = 9.2 \text{ cm}$ correct to 2 significant figures. Side of square = 4.1 cm correct to 2 significant figures. By considering bounds, work out the value of X to a suitable degree of accuracy. Show your working clearly. $X =$

YOUR SOLUTION

02 Find lower and upper bounds

TRIGGER *"work out", "find", a missing value*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.25 \leq w < 4.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q11

Estimate the value

$$4.25 \leq w < 4.35$$

Find upper bound

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) 4.35 kg, (b) 4.25 kg, (c) 8.15

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

11 The weight of a cat is 4.3 kg correct to 2 significant figures. (a) Write down the upper bound of the weight of the cat. kg (1) (b) Write down the lower bound of the weight of the cat. kg (1) G = e - f e = 17 correct to the nearest integer f = 9.4 correct to one decimal place (c) Work out the upper bound for the value of G. (2)

YOUR SOLUTION

03 Choose bounds for largest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Choose bounds for smallest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

16 $P = 2a c d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

06 Find upper bound

TRIGGER *"work out", "find", a missing value*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Round to the requested accuracy
"work out", "find", a missing value	Find lower and upper bounds
"work out", "show", "hence", a familiar exam phrase	Choose bounds for largest value
"work out", "show", "hence", a familiar exam phrase	Choose bounds for smallest value
"work out", "show", "hence", a familiar exam phrase	Estimate the value
"work out", "find", a missing value	Find upper bound
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Surds

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SURDS · CH1 · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Surds into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Simplify by square factors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Add and subtract like surds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Expand surd brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Rationalise the denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Simplify by square factors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q23

The volume is

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

Simplify surd

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

Finish: $2 + \sqrt{5}$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

23 The diagram shows a cuboid with a square cross section. x cm x cm $(25 - 3)$ cm
Diagram NOT accurately drawn The volume of the cuboid is $(13 + 65)$ cm³ Without using a calculator, find the value of x Give your answer in the form $a + b$ where a and b are integers. Show your working clearly. $x =$

YOUR SOLUTION

02 Add and subtract like surds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Expand surd brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q18

Simplify surd

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

Simplify surd

$$y\sqrt{x} + 21$$

Finish: $x = 19, y = 3$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Rationalise the denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2(2\sqrt{5} - 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q23

The cuboid has square cross

$$x^2(2\sqrt{5} - 3)$$

Given

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

Finish: $2 + \sqrt{5}$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

23 The diagram shows a cuboid with a square cross section. x cm x cm $(25 - 3)$ cm
Diagram NOT accurately drawn The volume of the cuboid is $(13 + 65)$ cm³ Without using a calculator, find the value of x Give your answer in the form $a + b$ where a and b are integers. Show your working clearly. $x =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Simplify by square factors
"work out", "show", "hence", a familiar exam phrase	Add and subtract like surds
"work out", "show", "hence", a familiar exam phrase	Expand surd brackets
"work out", "show", "hence", a familiar exam phrase	Rationalise the denominator
"work out", "show", "hence", a familiar exam phrase	Simplify surd
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Using a Calculator

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

USING A CALCULATOR · CH1 · L11

ENG. ESLAM AHMED



The Map

This lesson collapses Using a Calculator into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Type the whole expression safely

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use brackets for fractions and powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Read standard form display

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Round only at the end

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 67 website questions, 67 checked solutions, 3 local PDFs read.

01 Type the whole expression safely

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.06 \times 10^8 = 106000000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q10

Atlantic surface area

$$1.06 \times 10^8 = 106000000$$

Indian surface area

$$6.86 \times 10^7 = 68600000$$

Finish: (a) 3.74×10^7 , (b) 11

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean. Give your answer in standard form. square kilometres

(2) The surface area of the Pacific Ocean is k times the surface area of the ...

YOUR SOLUTION

02 Use brackets for fractions and powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

03 Read standard form display

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \frac{6}{10} \times 7 \frac{10}{10} \times 2 \frac{8}{10} \times 4 \frac{3}{10}$. . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

04 Round only at the end

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

06 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Type the whole expression safely
"work out", "show", "hence", a familiar exam phrase	Use brackets for fractions and powers
"work out", "show", "hence", a familiar exam phrase	Read standard form display
"work out", "show", "hence", a familiar exam phrase	Round only at the end
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	Estimate the value
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Ratio Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RATIO TOOLKIT · CH1 · L12



ENG. ESLAM AHMED

The Map

This lesson collapses Ratio Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find one share

TRIGGER *"work out", "find", a missing value*

02 Use total parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Convert mixed information to parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check the order of the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Split the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Total sailing

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 5 website questions, 5 checked solutions, 3 local PDFs read.

01 Find one share

TRIGGER *"work out", "find", a missing value*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

02 Use total parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

03 Convert mixed information to parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 \text{ parts} = 36$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2021 P1H Q4

Ducks pigs

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$9 \text{ parts} = 36$$

Finish: 70

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 On a farm there are chickens, ducks and pigs. The ratio of the number of chickens to the number of ducks is $7 : 2$ The ratio of the number of ducks to the number of pigs is $5 : 9$ There are 36 pigs on the farm. Work out the number of chickens on the farm.

YOUR SOLUTION

04 Check the order of the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

05 Split the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

06 Total sailing

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find one share
"work out", "show", "hence", a familiar exam phrase	Use total parts
"work out", "show", "hence", a familiar exam phrase	Convert mixed information to parts
"work out", "show", "hence", a familiar exam phrase	Check the order of the ratio
"work out", "show", "hence", a familiar exam phrase	Split the ratio
"work out", "show", "hence", a familiar exam phrase	Total sailing

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Ratio Problem Solving

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RATIO PROBLEM SOLVING · CH1 · L13



ENG. ESLAM AHMED

The Map

This lesson collapses Ratio Problem Solving into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find one share

TRIGGER *"work out", "find", a missing value*

02 Use total parts

TRIGGER *"probability", "random", "given that"*

03 Convert mixed information to parts

TRIGGER *"probability", "random", "given that"*

04 Check the order of the ratio

TRIGGER *"probability", "random", "given that"*

05 Split the ratio

TRIGGER *"probability", "random", "given that"*

06 The ratio is

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 7 website questions, 7 checked solutions, 3 local PDFs read.

01 Find one share

TRIGGER *"work out", "find", a missing value*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6 The total amount of money that Asha and Julie receive is \$36 Work out the amount of money that Brendon receives. \$ 1 5 2

YOUR SOLUTION

02 Use total parts

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Convert mixed information to parts

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Danil : Gabriel : Hadley} = 3 : 5 : 9$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q2

The ratio is

$$\text{Danil : Gabriel : Hadley} = 3 : 5 : 9$$

Split the ratio

$$9 - 5 = 4 \text{ parts}$$

Finish: Danil receives 147 euros.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

2 Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9 The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros. Work out the amount of money that Danil receives. euros

YOUR SOLUTION

04 Check the order of the ratio

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Split the ratio

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{7}{10} \times 20160 = 14112$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q1
of the money is changed

$$\frac{7}{10} \times 20160 = 14112$$

Split the ratio

$$\text{dollars} : \text{rupees} : \text{francs} = 9 : 5 : 2$$

Finish: 6174 rand more is changed into dollars than francs.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

1 Pieter owns a currency conversion shop. Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies. He changed 3 10 of the 20 160 rand into euros. He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2 Pieter changed more rands into dollars than he changed into francs. Work out how many more. rand

YOUR SOLUTION

06 The ratio is

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6 The total amount of money that Asha and Julie receive is \$36 Work out the amount of money that Brendon receives. \$ 1 5 2

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find one share
"probability", "random", "given that"	Use total parts
"probability", "random", "given that"	Convert mixed information to parts
"probability", "random", "given that"	Check the order of the ratio
"probability", "random", "given that"	Split the ratio
"probability", "random", "given that"	The ratio is

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Exchange Rates & Best Buys

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

EXCHANGE RATES & BEST BUYS · CH1 · L14



ENG. ESLAM AHMED

The Map

This lesson collapses Exchange Rates & Best Buys into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Write the exchange direction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Convert to one currency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Compare unit prices

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Round money sensibly

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 litres

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 1 website questions, 1 checked solutions, 3 local PDFs read.

01 Write the exchange direction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

02 Convert to one currency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

03 Compare unit prices

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

04 Round money sensibly

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

06 litres

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Write the exchange direction
"work out", "show", "hence", a familiar exam phrase	Convert to one currency
"work out", "show", "hence", a familiar exam phrase	Compare unit prices
"work out", "show", "hence", a familiar exam phrase	Round money sensibly
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	litres

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Direct & Inverse Proportion

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

DIRECT & INVERSE PROPORTION · CH1 · L15

ENG. ESLAM AHMED



The Map

This lesson collapses Direct & Inverse Proportion into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Choose direct or inverse

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find the constant k

TRIGGER *"work out", "find", a missing value*

03 Substitute the new value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check the direction of change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find inverse function

TRIGGER *"work out", "find", a missing value*

06 Use when

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 16 website questions, 16 checked solutions, 3 local PDFs read.

01 Choose direct or inverse

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

15 A is inversely proportional to C
A = 40 when C = 1.5
Calculate the value of C when A = 1000
C =

YOUR SOLUTION

02 Find the constant k

TRIGGER *"work out", "find", a missing value*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y \propto \frac{1}{\sqrt{x}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q20

Find inverse function

$$y \propto \frac{1}{\sqrt{x}}$$

Find inverse function

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $T = \frac{100}{9}$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

20 y is inversely proportional to x x is directly proportional to T 3 Given that y = 8 when T = 25 find the exact value of T when y = 27 T =

YOUR SOLUTION

03 Substitute the new value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

15 A is inversely proportional to C
A = 40 when C = 1.5
Calculate the value of C when A = 1000
C =

YOUR SOLUTION

04 Check the direction of change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$M = kh^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q17

Find inverse function

$$M = kh^3$$

Use when

$$4 = k(0.5)^3$$

Finish: $h = 2.5$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 M varies directly as the cube of h M = 4 when h = 0.5 Find the value of h when M = 500

YOUR SOLUTION

05 Find inverse function

TRIGGER *"work out", "find", a missing value*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Use when

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Choose direct or inverse
"work out", "find", a missing value	Find the constant k
"work out", "show", "hence", a familiar exam phrase	Substitute the new value
"work out", "show", "hence", a familiar exam phrase	Check the direction of change
"work out", "find", a missing value	Find inverse function
"work out", "show", "hence", a familiar exam phrase	Use when

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebra Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRA TOOLKIT · CH2 · L1



ENG. ESLAM AHMED

The Map

This lesson collapses Algebra Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 106 website questions, 106 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q1

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

Finish: (a) x^7 , (b) 7^9 .

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

1 (a) Simplify $x \times 9^2$ (1) (b) Write $7^7 7^8 4^3$ times as a single power of 7 (2)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2023 P2H Q12

(a)

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

Use index laws

$$4^x = (2^2)^x = 2^{2x}$$

Finish: (a) $x = \frac{17}{2}$, (b) $25p^4y^{16}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 2 4 4 7 5 x (a) Calculate the value of x x = (2) (b) Simplify fully $125^6 24^2 3^p y^2$

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 (a) Express $\frac{4}{2} - \frac{3}{1} \times x - - +$ as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify $2x(x - 5)(x - 3)$ (3)

YOUR SOLUTION

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

21 (a) Simplify fully $10^3 23 12 4 9 2 2 \times x \times x + + - (3) 2 2 8 4 2 3 2 5 y y y n$ times = + (b) Find an expression for n in terms of y . Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

21 (a) Simplify fully $10^2 3^{12} 4^9 2^2 \times x \times x + + - (3)^2 2^8 4^2 3^2 5^y y y n$ times = + (b)
Find an expression for n in terms of y. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Factorise expression
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Roots & Indices

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC ROOTS & INDICES · CH2 · L2



ENG. ESLAM AHMED

The Map

This lesson collapses Algebraic Roots & Indices into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use index laws with letters

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use negative powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Simplify before substituting

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 49 website questions, 49 checked solutions, 3 local PDFs read.

01 Use index laws with letters

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{12}{13} \times 4 \frac{3}{2} + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{12}{15} \times 2 \frac{3}{4} \div 3$ a t - (3)

YOUR SOLUTION

02 Use negative powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q1

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

Finish: (a) x^7 , (b) 7^9 .

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

21 (a) Simplify fully $10^2 3^{12} 4^9 2^2 \times x \times x + + - (3)^2 2^8 4^2 3^2 5^y y y n$ times = + (b)
Find an expression for n in terms of y. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

04 Simplify before substituting

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{1}{2} \times 7 \frac{4}{5} \times 3 \frac{2}{3} + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{1}{2} \times 15 \frac{2}{3} \times 3 \frac{1}{4} \times 2 \frac{3}{4}$ a t - (3)

YOUR SOLUTION

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

21 (a) Simplify fully $10x^3 - 12x^2 + 4x - 9$ $2x^3 + 2x^2 + 8x + 4$ $2x^3 + 3x^2 + 5x - n$ times = + (b)
Find an expression for n in terms of y . Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

06 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Roots & Indices question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{12} = 2\sqrt{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q17

Simplify surd

$$\sqrt{12} = 2\sqrt{3}$$

Simplify surd

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

Finish: (a) shown, (b) $\frac{t^{10}}{9a^8}$

FORMULA BANK

Use index laws before expanding. Fractional index means root; negative index means reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

17 (a) Show that $6 \frac{2}{3} \times 12 \frac{12}{13} \times 7 \frac{4}{5} \times 3 \frac{2}{3} + () = + ()$ Show each stage of your working. (3) (b) Simplify fully $27 \frac{12}{15} \times 2 \frac{3}{4} \times 1 - (3)$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use index laws with letters
"work out", "show", "hence", a familiar exam phrase	Use negative powers
"work out", "show", "hence", a familiar exam phrase	Use fractional powers
"work out", "show", "hence", a familiar exam phrase	Simplify before substituting
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Simplify surd

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Expanding Brackets

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

EXPANDING BRACKETS · CH2 · L3

ENG. ESLAM AHMED



The Map

This lesson collapses Expanding Brackets into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Expand brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 20 website questions, 20 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 (a) Express $\frac{4}{2} - \frac{3}{1} \times x - - +$ as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify $2x(x - 5)(x - 3)$ (3)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P2H Q12

(a)

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

(b)

$$\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x} = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

Finish: (a) $16p^6q^8$, (b) $\frac{17}{30x}$, (c) $8x^3 - 28x^2 - 60x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

12 (a) Simplify $\frac{64p^9q^{12}}{2^3 \cdot 2}$ (b) Write as a single fraction $\frac{2^3 \cdot 4^5 \cdot 9^{10}}{x \cdot x \cdot x} - \frac{1}{x}$. Give your answer in its simplest form. (2) (c) Expand and simplify $4x(x - 5)(2x + 3)$. Show your working clearly. (3)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q1

Expand brackets

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

Finish: 28, $2p(4p - 1)$, $12t^2 - 8t$, $5x^2 + 18x - 8$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

1 $w = 5y^2 - y^3$ (a) Work out the value of w when $y = -2$ $w = (2)$ (b) Factorise fully $8p^2 - 2p$ (2) (c) Expand $4t(3t - 2)$ (2) (d) Expand and simplify $(5x - 2)(x + 4)$ (2)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 (a) Express $4\frac{2}{3} - 1\frac{1}{x} + \frac{1}{x}$ as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify $2x(x - 5)(x - 3)$ (3)

YOUR SOLUTION

05 Expand brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q1

Expand brackets

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

Finish: 28, $2p(4p - 1)$, $12t^2 - 8t$, $5x^2 + 18x - 8$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

1 $w = 5y^2 - y^3$ (a) Work out the value of w when $y = -2$ $w = (2)$ (b) Factorise fully $8p^2 - 2p$ (2) (c) Expand $4t(3t - 2)$ (2) (d) Expand and simplify $(5x - 2)(x + 4)$ (2)

YOUR SOLUTION

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2HR Q13

(a)

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

(b)

$$4x(x + 3)(2x - 5)$$

Finish: (a) $(2x - 5y)(2x + 5y)$, (b) $8x^3 + 4x^2 - 60x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

13 (a) Factorise $4x^2y - 2xy^2$. (2) (b) Show that $4x^3 + 5x^2 + 3x + 2$ can be written in the form $ax^3 + bx^2 + cx + 2$ where a , b and c are integers to be found. (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Expand brackets
"work out", "show", "hence", a familiar exam phrase	Factorise expression

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Factorising

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FACTORISING · CH2 · L4

ENG. ESLAM AHMED



The Map

This lesson collapses Factorising into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Take out the common factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Factorise a quadratic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use difference of squares

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check by expanding

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 (b)(i)

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 23 website questions, 23 checked solutions, 3 local PDFs read.

01 Take out the common factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Factorise a quadratic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use difference of squares

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$n^2 - 7n + 12 = (n - 3)(n - 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q9

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) x^9 , (b) $64y^6$, (c) $(n - 3)(n - 4)$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

9 (a) Simplify x^4 times x^5 (1) (b) Simplify $(4y^2)^3$ (2) (c) Factorise $n^2 - 7n + 12$ (2)

04 Check by expanding

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q11

(a)

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

(b)

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

Finish: (a) $(3x - 2y)(3x + 2y)$, (b) $\frac{5x-6}{8x}$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 (b)(i)

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Factorising question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q9

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

Finish: (a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

FORMULA BANK

Look for common factor first, then brackets, then difference of squares or quadratic factorising.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Take out the common factor
"work out", "show", "hence", a familiar exam phrase	Factorise a quadratic
"work out", "show", "hence", a familiar exam phrase	Use difference of squares
"work out", "show", "hence", a familiar exam phrase	Check by expanding
"work out", "show", "hence", a familiar exam phrase	Factorise expression
"work out", "show", "hence", a familiar exam phrase	(b)(i)

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Completing the Square

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMPLETING THE SQUARE · CH2 · L5



ENG. ESLAM AHMED

The Map

This lesson collapses Completing the Square into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Halve the coefficient of x

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Add and subtract the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Find turning point form

TRIGGER *"work out", "find", a missing value*

04 Solve using completed square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Halve the coefficient of x

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Add and subtract the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

YOUR SOLUTION

03 Find turning point form

TRIGGER *"work out", "find", a missing value*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2H Q25

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

(b)

$$y = 52 - 6(x - 2)^2$$

Finish: (a) $52 - 6(x - 2)^2$. For the sketch, show a downward parabola with turning point (2, 52) and y -intercept (0, 28).

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

25 (a) Write $28x^2 - 6x + x^3$ in the form $a(x - b)^2 + c$ where a , b and c are integers. (3) (b) On the axes below, sketch the graph of $y = 28x^2 - 6x + x^3$. Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis. (3)

YOUR SOLUTION

04 Solve using completed square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q19

(a)

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

Evaluate fraction

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $m = -\frac{1}{3}$, and $3(x+2)^2 + 7$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

19 Given that $13 \times m$ (a) find the value of m $m = (1)$ Given that a, b and c are integers, (b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$ (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Halve the coefficient of x
"work out", "show", "hence", a familiar exam phrase	Add and subtract the square
"work out", "find", a missing value	Find turning point form
"work out", "show", "hence", a familiar exam phrase	Solve using completed square
"work out", "show", "hence", a familiar exam phrase	Evaluate fraction
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Fractions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC FRACTIONS · CH2 · L6



ENG. ESLAM AHMED

The Map

This lesson collapses Algebraic Fractions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Factorise before cancelling

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use common denominators

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Multiply and divide algebraic fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 State excluded values when needed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Factorise before cancelling

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1H Q12

(a)

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

(b)

$$\frac{2x + 1}{4x} + \frac{7 - 5x}{3x}$$

Finish: (a) $(3y - 4)(2y + 1)$, (b) $\frac{31 - 14x}{12x}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 (a) Factorise fully $6y^2 - 5y - 4$ (2) (b) Express $2\frac{1}{4} \times 7\frac{5}{3} \times x \times x \times x$ as a single fraction in its simplest form. (3)

YOUR SOLUTION

02 Use common denominators

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish: $\boxed{\frac{1-3x}{x(2x-3)^2}}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

24 Express $4\frac{2}{5} - 3\frac{2}{3} + 9\frac{4}{6} - 17\frac{5}{3} + 2x + x + x + x + x$ divided by $- - +$ as a single fraction in its simplest form.

YOUR SOLUTION

03 Multiply and divide algebraic fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish:

$$\frac{1-3x}{x(2x-3)^2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

24 Express $\frac{4 \cdot 2 \cdot 5 \cdot 3 \cdot 2 \cdot 3 \cdot 9 \cdot 4 \cdot 6 \cdot 17 \cdot 5 \cdot 3 \cdot 2 \cdot x \cdot x \cdot x \cdot x \cdot x}{- - +}$ divided by $- - +$ as a single fraction in its simplest form.

YOUR SOLUTION

04 State excluded values when needed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q21

Use trigonometry

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

Factor

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

Finish:

$$\frac{54}{5x + 2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

21 Write $\frac{25}{64} \times \frac{5}{13} \times \frac{6}{8} \times \frac{15}{5} \times \frac{8}{7} \times \frac{2}{2} \times x \times x \times x \times x \times x$ --- times - + + - - () as a single fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q24

Use trigonometry

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

Use trigonometry

$$\frac{4}{2x-5} - \frac{3}{2x-3} = \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)}$$

Finish:

$$\frac{1-3x}{x(2x-3)^2}$$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

24 Express $4\frac{2}{5} - 3\frac{2}{3} + 9\frac{4}{6} - 7\frac{5}{3} - 2x + x + x + x + x$ divided by $- +$ as a single fraction in its simplest form.

YOUR SOLUTION

06 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q19

Simplify fraction

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

Simplify fraction

$$3x-7 = \frac{21}{4y-3} - 7$$

Finish: $a = \frac{4y-3}{3-2y}$

FORMULA BANK

Factorise first, cancel common factors, then use common denominator if adding.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

19 a = 14 3 7 x - x = 7 4 3 y - Express a in the form py q ry s + + where p, q, r and s are integers. Give your answer in its simplest form. a =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Factorise before cancelling
"work out", "show", "hence", a familiar exam phrase	Use common denominators
"work out", "show", "hence", a familiar exam phrase	Multiply and divide algebraic fractions
"work out", "show", "hence", a familiar exam phrase	State excluded values when needed
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Simplify fraction

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Rearranging Formulas

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

REARRANGING FORMULAS · CH₂ · L7



ENG. ESLAM AHMED

The Map

This lesson collapses Rearranging Formulas into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Make the subject visible

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Clear fractions first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Undo operations in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Factor out the subject

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Rearrange formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Square both sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 19 website questions, 19 checked solutions, 3 local PDFs read.

01 Make the subject visible

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$d = g + 2ac$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q3

(a)

$$d = g + 2ac$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

Finish: $a = \frac{d-g}{2c}, 3f(3e-4), x^2 - 3x - 10, n^6$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 (a) Make a the subject of $d = g + 2ac$ (2) (b) Factorise fully $9e^2f - 12ef^2$ (2) (c) Expand and simplify $(x + 2)(x - 5)$ (2) (d) Simplify fully $n^4n^7n^5$ (2)

YOUR SOLUTION

02 Clear fractions first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q14

(a)

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

Multiply by 20

$$4(9a - 7) - 5(3a - 7) = 91$$

Finish: (a) $a = 4$, (b) $c = \frac{8-3p^2}{p^2-a}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 (a) Solve $9.75374a - 4.55 = 4.55$. Show clear algebraic working. $a = 3$ (b) Make c the subject of the formula $p = ac + 83$ (4)

YOUR SOLUTION

03 Undo operations in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{c}{y} - 5z$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q6

(a)

$$A = \frac{c}{y} - 5z$$

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $c = y(A + 5z)$, 1, $(x - 3)(x - 8)$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Make c the subject of $A = c y - 5z$ (2) (b) Write down the value of g 0 (1) (c) Factorise $x^2 - 11x + 24$ (2)

YOUR SOLUTION

04 Factor out the subject

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$d = g + 2ac$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q3

(a)

$$d = g + 2ac$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

Finish: $a = \frac{d-g}{2c}, 3f(3e-4), x^2 - 3x - 10, n^6$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

- 3 (a) Make a the subject of $d = g + 2ac$ (2) (b) Factorise fully $9e^2f - 12ef^2$ (2) (c) Expand and simplify $(x + 2)(x - 5)$ (2) (d) Simplify fully $n^4n^7n^5$ (2)

YOUR SOLUTION

05 Rearrange formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = \sqrt{\frac{x+1}{x-4}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q16

Rearrange formula

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides

$$y^2 = \frac{x+1}{x-4}$$

Finish: $x = \frac{4y^2+1}{y^2-1}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

16 Make x the subject of $y = x^2 + -14$

YOUR SOLUTION

06 Square both sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = \sqrt{\frac{x+1}{x-4}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q16

Rearrange formula

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides

$$y^2 = \frac{x+1}{x-4}$$

Finish: $x = \frac{4y^2+1}{y^2-1}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Make the subject visible
"work out", "show", "hence", a familiar exam phrase	Clear fractions first
"work out", "show", "hence", a familiar exam phrase	Undo operations in order
"work out", "show", "hence", a familiar exam phrase	Factor out the subject
"work out", "show", "hence", a familiar exam phrase	Rearrange formula
"work out", "show", "hence", a familiar exam phrase	Square both sides
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebraic Proof

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRAIC PROOF · CH2 · L8



ENG. ESLAM AHMED

The Map

This lesson collapses Algebraic Proof into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Start from the expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Expand and collect

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use parity or multiple language

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Finish with the required statement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Start from the expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x = 0.5727272 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q13

(a) Let

$$x = 0.5727272 \dots$$

Calculate value

$$1000x = 572.727272 \dots$$

Finish: (a) shown, (b) $\frac{6+3\sqrt{y}}{4-y}$

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

13 (a) Use algebra to show that $0.57 \cdot 2 = 63 \cdot 110$ (2) Given that y is a prime number, (b) express $3 \cdot 2 - y$ in the form $a \cdot b \cdot y \cdot c \cdot y + \dots$ where a, b and c are integers. (2)

YOUR SOLUTION

02 Expand and collect

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q17

Let the two consecutive square

$$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

is one more than an

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The difference is always odd.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use parity or multiple language

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$N = 5k$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q25

Use trigonometry

$$N = 5k$$

where is an integer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $A^2 - B^2$ is always a multiple of 20.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

25 N is a multiple of 5 $A = N + 1$ $B = N - 1$ Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

YOUR SOLUTION

04 Finish with the required statement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$N = 5k$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q25

Use trigonometry

$$N = 5k$$

where is an integer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $A^2 - B^2$ is always a multiple of 20.

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

25 N is a multiple of 5 $A = N + 1$ $B = N - 1$ Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

YOUR SOLUTION

05 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebraic Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x = 0.5727272 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q13

(a) Let

$$x = 0.5727272 \dots$$

Calculate value

$$1000x = 572.727272 \dots$$

Finish: (a) shown, (b) $\frac{6+3\sqrt{y}}{4-y}$

FORMULA BANK

Start with an expression, expand/simplify until the required form appears.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

13 (a) Use algebra to show that $0.57 \cdot 2 = 63 \cdot 110$ (2) Given that y is a prime number, (b) express $3 \cdot 2 - y$ in the form $a \cdot b \cdot y \cdot c \cdot y + -$ where a, b and c are integers. (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Start from the expression
"work out", "show", "hence", a familiar exam phrase	Expand and collect
"work out", "show", "hence", a familiar exam phrase	Use parity or multiple language
"work out", "show", "hence", a familiar exam phrase	Finish with the required statement
"work out", "show", "hence", a familiar exam phrase	Subtract

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Linear Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING LINEAR EQUATIONS · CH2 · L9



ENG. ESLAM AHMED

The Map

This lesson collapses Solving Linear Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x - 5) = \frac{9 - x}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1 + 4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

7 (a) Expand and simplify $(m - 8)(m + 5)$ (2) (b) Factorise fully $5y + 20y^2$ (2) (c) Simplify $(p^2 + 3)^0$ (1) (d) Solve $3(2x - 5) = 9$ 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(n - 6)(n + 4) = n^2 - 2n - 24$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q5

(a)

$$(n - 6)(n + 4) = n^2 - 2n - 24$$

(b)

$$2x - 3 = \frac{3x - 5}{4}$$

Finish: (a) $n^2 - 2n - 24$, (b) $x = \frac{7}{5}$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 (a) Expand and simplify $(n - 6)(n + 4)$ (2) (b) Solve $2x - 3 = 3$ 5 4 x - Show clear algebraic working. $x = (3)$

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{5a + 8}{3} - \frac{2a + 5}{4} = 23$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P1HR Q15

Solve equation

$$\frac{5a + 8}{3} - \frac{2a + 5}{4} = 23$$

Multiply by

$$4(5a + 8) - 3(2a + 5) = 276$$

Finish: (a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

15 (a) Solve $5^8 3^2 5^4 2^3$ a Show clear algebraic working. a = (4) (b) Express y^3 in the form $c y^n$ where c and n are numbers to be found. (2)

YOUR SOLUTION

06 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Linear Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q15

(a)

$$\frac{4x+5}{3} - \frac{3-2x}{2} = 13$$

Multiply by 6

$$2(4x+5) - 3(3-2x) = 78$$

Finish: $x = \frac{11}{2}$, and $-\frac{5}{2} \leq y \leq 6$.

FORMULA BANK

Expand, collect x -terms on one side, numbers on the other, then divide.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

15 (a) Solve $4.5 - 3 + 2x - 2 = 13$ Show clear algebraic working. $x = (4)$ (b) Solve the inequality $2y^2 - 7y - 30 < 0$ Show your working clearly. (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "show", "hence", a familiar exam phrase	Solve quadratic equation

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Quadratic Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING QUADRATIC EQUATIONS · CH2 · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Solving Quadratic Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Factorise then solve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the quadratic formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Complete the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Use graph roots

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 4 website questions, 4 checked solutions, 3 local PDFs read.

01 Factorise then solve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use the quadratic formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Complete the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 21x + 20 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q9

Solve quadratic equation

$$x^2 - 21x + 20 = 0$$

Factorise

$$(x - 20)(x - 1) = 0$$

Finish: $x = 20$ or $x = 1$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Use graph roots

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5x^2 + 9x - 17 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1H Q15

Solve quadratic equation

$$5x^2 + 9x - 17 = 0$$

Use quadratic formula

$$x = \frac{-9 \pm \sqrt{9^2 - 4(5)(-17)}}{2(5)}$$

Finish: $x = 1.15$ or $x = -2.95$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Solve quadratic equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Quadratic Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2 - 5x - 36 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q6

Solve quadratic equation

$$x^2 - 5x - 36 = 0$$

Factorise

$$(x - 9)(x + 4) = 0$$

Finish: $x = 9$ or $x = -4$.

FORMULA BANK

Factorise, complete square, or use formula. Each bracket equal to zero gives a root.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Factorise then solve
"work out", "show", "hence", a familiar exam phrase	Use the quadratic formula
"work out", "show", "hence", a familiar exam phrase	Complete the square
"work out", "show", "hence", a familiar exam phrase	Use graph roots
"work out", "show", "hence", a familiar exam phrase	Solve quadratic equation
"work out", "show", "hence", a familiar exam phrase	Factorise

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Solving Inequalities

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SOLVING INEQUALITIES · CH2 · L11



ENG. ESLAM AHMED

The Map

This lesson collapses Solving Inequalities into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve inequality

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 22 website questions, 22 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q19

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

Multiply by 10

$$2(4-3x) - 5(3x-5) = -30$$

Finish: $x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

19 (a) Solve $4^3 5^3 5^2 3^{---} = -x^x$ Show clear algebraic working. $x = (3)$ (b) Solve the inequality $5y^2 - 17y < 40$ (3)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq y < 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q3

Solve inequality

$$-2 \leq y < 1$$

use a closed circle at

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Closed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

3 (a) On the number line, show the inequality $-2 < y < 1$ (2) (b) Write down all the values of n that satisfy $-3.4 < n < 2$ (2)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q5

(a)

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

Expand brackets

$$\frac{y^5 \times y^n}{y^6} = y^{n-1} = y^{13}$$

Finish: (a) $2x^2 + 29x$, (b) $n = 14$, (c) $t < 3$

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

5 (a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$ (2) Given that $y \times y \times y \times n \times 5 \times 6$ times = y^{13}
 (b) work out the value of n . $n = (2)$ (c) (i) Solve the inequality $7t - 8 < 2t + 7$ (2) (ii) On
 the number line below, represent the solution set of the inequality solved in part (c)(i)
 $-5 -4 -3 -2 -1 0 1 2 3 4 5 t$ (1)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq n < 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q1

(a)

$$-2 \leq n < 3$$

Calculate value

$$n = -2, -1, 0, 1, 2$$

Finish: $n = -2, -1, 0, 1, 2$; closed at 1, shaded left.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

1 n is an integer. (a) Write down all the values of n such that $-2 < n < 3$ (2) (b) On the number line, represent the inequality $y < 1$ -3 -2 -1 0 1 2 3 y (1)

YOUR SOLUTION

05 Solve inequality

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$-2 \leq y < 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q3

Solve inequality

$$-2 \leq y < 1$$

use a closed circle at

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Closed at -2 , open at 1 ; integers $-3, -2, -1, 0, 1, 2$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 (a) On the number line, show the inequality $-2 < y < 1$ (2) (b) Write down all the values of n that satisfy $-3.4 < n < 2$ (2)

YOUR SOLUTION

06 Factorise

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Solving Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q19

(a)

$$\frac{4-3x}{5} - \frac{3x-5}{2} = -3$$

Multiply by 10

$$2(4-3x) - 5(3x-5) = -30$$

Finish: $x = 3$, and $-\frac{8}{5} \leq y \leq 5$.

FORMULA BANK

Solve like an equation, but flip the sign when multiplying/dividing by a negative.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

19 (a) Solve $4^3 5^3 5^2 3^{---} = -x^x$ Show clear algebraic working. $x = (3)$ (b) Solve the inequality $5y^2 - 17y < 40$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Solve inequality
"work out", "show", "hence", a familiar exam phrase	Factorise

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Simultaneous Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SIMULTANEOUS EQUATIONS · CH2 · L12



ENG. ESLAM AHMED

The Map

This lesson collapses Simultaneous Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve simultaneous equations

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

| Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x + 7y = 17$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q12

Solve simultaneous equations

$$2x + 7y = 17$$

Multiply the first equation by

$$10x + 35y = 85$$

Finish: $x = -2, y = 3$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

| Solve $3x + 2y = 17$ and $2x - y = 4$.

YOUR SOLUTION

05 Solve simultaneous equations

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Simultaneous Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3x + 5y = 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q7

Solve simultaneous equations

$$3x + 5y = 6$$

Add the equations

$$10x = -5$$

Finish: $x = -\frac{1}{2}, y = \frac{3}{2}$.

FORMULA BANK

Eliminate one letter or substitute, then solve the remaining equation.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve simultaneous equations

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Forming & Solving Equations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FORMING & SOLVING EQUATIONS · CH2 · L13

ENG. ESLAM AHMED



The Map

This lesson collapses Forming & Solving Equations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 7 website questions, 7 checked solutions, 3 local PDFs read.

01 Collect like terms

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

02 Solve with brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x + 6 = 30 - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1HR Q6

Use trigonometry

$$4x + 6 = 30 - x$$

Finish: $x = 4.8$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 6 2 10 p (c) Find the value of p p = (1) 5 1 r = (d) Write down the value of r r = (1) 6
The diagram shows triangle ABC A B C Diagram NOT accurately drawn (4x + 6) cm
(30 - x) cm CA x 4 6 cm CB x 30 cm angle angle CAB CBA = Work out the value of x x
=

YOUR SOLUTION

03 Form the equation from words

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

04 Check in the original

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4(5 - x)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P2H Q2

Area of rectangle A

$$4(5 - x)$$

Area of rectangle B

$$5(2x - 1)$$

Finish: $x = 2.5$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

2 Here are two rectangles, rectangle A and rectangle B. Diagram NOT accurately drawn

4 cm	5 cm
(5 - x) cm	(2x - 1) cm

rectangle B rectangle A

The area of rectangle B is twice the area of rectangle A. Work out the value of x. Show your working clearly.

 $x =$

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x + 4x + (x - 7) = 137$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q3

Why it moved The question

This is a reasoning step: write one clear sentence, then continue the calculation.

Let the number of buttons

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 17 buttons.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 Three tins, A, B and C, each contain buttons. Tin A contains x buttons. Tin B contains 4 times the number of buttons that tin A contains. Tin C contains 7 fewer buttons than tin A. The total number of buttons in the three tins is 137. Work out the number of buttons in tin C.

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Forming & Solving Equations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x + 6 = 30 - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1HR Q6

Use trigonometry

$$4x + 6 = 30 - x$$

Finish: $x = 4.8$.

FORMULA BANK

Name the unknown, translate the sentence/diagram into an equation, then solve and check.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

6 6 2 10 p (c) Find the value of p p = (1) 5 1 r = (d) Write down the value of r r = (1) 6
The diagram shows triangle ABC A B C Diagram NOT accurately drawn (4x + 6) cm
(30 - x) cm CA x 4 6 cm CB x 30 cm angle angle CAB CBA = Work out the value of x x
=

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Collect like terms
"work out", "show", "hence", a familiar exam phrase	Solve with brackets
"work out", "show", "hence", a familiar exam phrase	Form the equation from words
"work out", "show", "hence", a familiar exam phrase	Check in the original
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sequences

Read the trigger, name the data, choose the construction, write the bottom line.

SEQUENCES · CH₃ · L1

ENG. ESLAM AHMED



The Map

This lesson is not a list of formulas. It is six exam moves. The question starts when you recognise the words that tell you what the working must become.

01 The rule (n th term)

TRIGGER *"nth term", "expression in terms of n ", "the rule"*

02 A named term or membership test

TRIGGER *"20th term", "is 150 a term?", "which term is ...?"*

03 The sum

TRIGGER *"sum", "total", "altogether", "from the 50th to the 100th"*

04 Two clues $\rightarrow a$ and d

TRIGGER *two facts about the same sequence, for example u_{20} and S_{30}*

05 Letters inside first terms

TRIGGER *terms like $8p$, $7p - 3$, $4p + 2$*

06 Context and geometry

TRIGGER *savings, shelves, polygons, counterexamples*

BANK EVIDENCE 30 unique solved source questions were mined from the Sequences bank, worth 134 marks. The largest weight is arithmetic series and algebraic sequence terms.

01 The Rule — n th Term

TRIGGER “ n th term”, “expression in terms of n ”, first few terms

BECOMES A rule that jumps straight from the position n to the term u_n .

FIRST LINE TO WRITE

$$d = \text{second term} - \text{first term} \quad \text{then} \quad u_n = a + (n - 1)d$$

SIMPLEST STRATEGY

- 1 Read the first term as a .
- 2 Find the common difference d .
- 3 Write $u_n = a + (n - 1)d$.
- 4 Expand and tidy.
- 5 Check by putting $n = 1$.

WORKED MODEL

First four terms: 85, 79, 73, 67.

$$d = 79 - 85 = -6$$

$$u_n = 85 + (n - 1)(-6) = 85 - 6n + 6 = 91 - 6n$$

BOTTOM LINE

$$u_n = 91 - 6n$$

SAME IDEA Find the n th-term rule from the first few terms.

QUESTION

Here are the first four terms of an arithmetic sequence:

38, 31, 24, 17

Find an expression, in terms of n , for the n th term of the sequence.

YOUR SOLUTION

02 A Named Term or a Membership Test

TRIGGER “the 20th term”, “which term is ...?”, “is 150 a term?”

BECOMES Either substitute a position into the rule, or solve the rule for n .

FIRST LINE TO WRITE

named term: substitute n

membership: solve $u_n = \text{given value}$

SIMPLEST STRATEGY

- 1 If the question says “20th”, put $n = 20$.
- 2 If it asks whether a value is in the sequence, set u_n equal to that value.
- 3 Solve for n .
- 4 The value is a term only if n is a positive whole number.

WORKED MODEL

The n th term is $4n + 7$. Show a counterexample to the rule $5n + 4$.

$$5(3) + 4 = 19$$

But 19 also appears in the first sequence:

$$4(3) + 7 = 19$$

BOTTOM LINE

A suitable counterexample is 19.

SAME IDEA Substitute a position, or solve the rule for n .

QUESTION

The n th term of a sequence is $5n + 17$.

(a) Find the 12th term.

(b) Is 152 a term in this sequence? Show your working.

YOUR SOLUTION

03 The Sum — Arithmetic Series

TRIGGER “sum”, “total”, “altogether”, S_n , “from term r to term s ”

BECOMES A series question. You are adding terms, not finding one term.

FIRST LINE TO WRITE

$$S_n = \frac{n}{2}(2a + (n-1)d) \quad \text{or} \quad S = \frac{\text{number}}{2}(\text{first} + \text{last})$$

SIMPLEST STRATEGY

- 1 Decide whether it is first- n or a block from one position to another.
- 2 For a block, find the first term of the block and the last term of the block.
- 3 Count the terms carefully: last position minus first position plus 1.
- 4 Use the first-plus-last version when both ends of the block are known.

WORKED MODEL

First terms: 8, 15, 22, 29, 36. Find the sum from the 50th term to the 100th term inclusive.

$$d = 7, \quad u_n = 8 + 7(n-1) = 7n + 1$$

$$u_{50} = 351, \quad u_{100} = 701, \quad \text{number} = 100 - 50 + 1 = 51$$

$$S = \frac{51}{2}(351 + 701) = 26826$$

— BOTTOM LINE

26826.

04 Two Clues $\rightarrow a$ and d

TRIGGER *two facts about the same arithmetic sequence, for example $u_{20} = 123$ and $S_{30} = 2880$*

BECOMES Two simultaneous equations in a and d .

FIRST LINE TO WRITE

$$u_r = a + (r - 1)d \quad \text{and} \quad S_n = \frac{n}{2}(2a + (n - 1)d)$$

SIMPLEST STRATEGY

- 1 Turn the term fact into equation 1.
- 2 Turn the sum fact into equation 2.
- 3 Simplify both equations.
- 4 Eliminate one letter.
- 5 Write both values before answering the final request.

WORKED MODEL

$$u_{20} = a + 19d = 123 \quad (1)$$

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$

$$2a + 29d = 192 \quad (2)$$

Double (1), then subtract (2):

$$2a + 38d = 246 \quad 9d = 54 \quad d = 6$$

$$a + 19(6) = 123 \quad a = 9$$

BOTTOM LINE

$$\boxed{a = 9, d = 6}.$$

SAME IDEA Turn two facts into two equations, then solve.

QUESTION

An arithmetic series has first term a and common difference d . The 21st term is 109. The sum of the first 52 terms is 4381. Work out the 5th term of the series.

YOUR SOLUTION

05 Letters Inside the First Terms

TRIGGER *terms such as $8p$, $7p - 3$, $4p + 2$*

BECOMES An equal-differences question first. The real sequence appears after the letter is found.

FIRST LINE TO WRITE

$$\text{term 2} - \text{term 1} = \text{term 3} - \text{term 2}$$

SIMPLEST STRATEGY

- 1 Use equal differences to solve the letter.
- 2 Replace the letter in the original terms.
- 3 Now identify a and d .
- 4 Continue with u_n or S_n , depending on what the question asks.

WORKED MODEL

The first three terms are $8p$, $7p - 3$, $4p + 2$.

$$(7p - 3) - 8p = (4p + 2) - (7p - 3)$$

$$-p - 3 = -3p + 5, \quad 2p = 8, \quad p = 4$$

So the sequence is:

$$32, \quad 25, \quad 18, \dots \quad a = 32, \quad d = -7$$

If $S_n = -1914$,

$$\frac{n}{2}(2(32) + (n - 1)(-7)) = -1914$$

$$\frac{n}{2}(71 - 7n) = -1914 \quad \Rightarrow \quad 7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 + 335}{14} = 29$$

BOTTOM LINE

$$\boxed{n = 29}.$$

SAME IDEA Use equal differences first, then continue with the real sequence.

QUESTION

The first three terms of an arithmetic sequence are

$$5q, \quad 4q + 3, \quad q + 12.$$

Find q . Then find the sum of the first 20 terms of the sequence.

YOUR SOLUTION

06 Context and Geometry

TRIGGER *savings by year, shelves, polygon angles, or one counterexample*

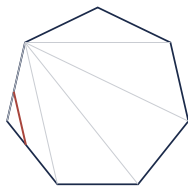
BECOMES A hidden sequence. Translate the story first, then use the same sequence rules.

FIRST LINE TO WRITE

name a, d, n then write the outside rule beside the sequence rule

SIMPLEST STRATEGY

- 1 Savings: count years first, especially when the years are inclusive.
- 2 Shelves or patterns: count each object type in terms of n .
- 3 Polygon angles: sequence sum equals $180(n - 2)$.
- 4 Counterexample: one valid value that breaks the claim is enough.



sequence of interior angles

$$\begin{array}{ccccccc}
 84^\circ & & 88^\circ & & 92^\circ & & \dots \\
 \bullet & \text{---} & \bullet & \text{---} & \bullet & \text{---} & \bullet \\
 & \text{---} & & \text{---} & & \text{---} &
 \end{array}$$

$$\begin{aligned}
 S_n &= \frac{n}{2}(2a + (n-1)d) \\
 S_n &= 180(n-2)
 \end{aligned}$$

FIGURE Match the angle-series sum with the polygon interior-angle sum.

WORKED MODEL

The interior angles of a polygon form an arithmetic sequence. The smallest angle is 84° , and the common difference is 4° .

$$S_n = \frac{n}{2}(2(84) + (n-1)4) = 2n^2 + 82n$$

The polygon interior sum is:

$$180(n-2)$$

Set equal:

$$2n^2 + 82n = 180(n-2)$$

$$n^2 - 49n + 180 = 0 \quad (n-45)(n-4) = 0$$

Since $n > 5$, $n = 45$, so

$$\text{sum} = 180(45-2) = 7740$$

BOTTOM LINE

7740°.

SAME IDEA Translate the story into a sequence rule and an outside rule.

QUESTION

The interior angles of a polygon form an arithmetic sequence. The smallest angle is 80° , and the common difference is 5° . The polygon has more than 5 sides. Work out the sum of the interior angles of the polygon.

YOUR SOLUTION

Quick Reference

TRIGGER → ACTION

IF YOU SEE	DO THIS FIRST
"nth term", "expression in terms of n "	$u_n = a + (n - 1)d$.
"the 20th term"	Substitute $n = 20$.
"is this value a term?"	Solve $u_n = \text{value}$ and check n is a positive integer.
"sum", "total", "altogether"	$S_n = \frac{n}{2}(2a + (n - 1)d)$.
"from term r to term s "	number of terms $= s - r + 1$.
two facts about the sequence	write two equations, then solve for a, d .
letters in first terms	equal differences first.
polygon angles	sequence sum $= 180(n - 2)$.

THE BOTTOM LINES

- 1 **Rule:** $u_n = 3n - 2$, or another clean expression in n .
- 2 **Term:** one value, such as $u_{20} = 77$.
- 3 **Membership:** the position n , or not a term.
- 4 **Sum:** one total, with units if the problem has units.
- 5 **Two clues:** a and d , then the exact thing asked for.
- 6 **Context:** the story answer, not only the algebra.

FORMULA BANK

$$u_n = a + (n - 1)d \quad d = u_2 - u_1 \quad S_n = \frac{n}{2}(2a + (n - 1)d) \quad S = \frac{\text{number}}{2}(\text{first} + \text{last})$$

$$180(n - 2)$$

PRACTICE ORDER

Linear nth term → block sums → two-equation questions → letters/context/geometry.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Coordinate Geometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COORDINATE GEOMETRY · CH₃ · L₂

ENG. ESLAM AHMED



The Map

This lesson collapses Coordinate Geometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use midpoint

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use gradient

TRIGGER *"gradient", "tangent", "rate of change"*

03 Find equation of a line

TRIGGER *"work out", "find", a missing value*

04 Check if a point lies on a line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 60 website questions, 60 checked solutions, 3 local PDFs read.

01 Use midpoint

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3 , and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

02 Use gradient

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3 , and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

03 Find equation of a line

TRIGGER *"work out", "find", a missing value*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: (9, -1.5), gradient -3, and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

04 Check if a point lies on a line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2} \right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3 , and no.

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2} \right)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q22

Find the midpoint

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2} \right)$$

Find the gradient

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

Finish: $j = -2, k = 11$

FORMULA BANK

Midpoint = $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$. Gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = 80$ H is the point with coordinates $(-4, 1)$ J is the point with coordinates $(j, 15)$ where $j < 0$ K is the point with coordinates $(6, k)$ M is the midpoint of JK. The gradient of HM is 2 Find the value of j and the value of k. $j = k =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use midpoint
"gradient", "tangent", "rate of change"	Use gradient
"work out", "find", a missing value	Find equation of a line
"work out", "show", "hence", a familiar exam phrase	Check if a point lies on a line
"work out", "find", a missing value	Find the gradient
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Functions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FUNCTIONS · CH3 · L3

ENG. ESLAM AHMED



The Map

This lesson collapses Functions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Substitute into a function

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find an inverse

TRIGGER *"work out", "find", a missing value*

03 Use composite functions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Solve from function notation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 31 website questions, 31 checked solutions, 3 local PDFs read.

01 Substitute into a function

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x + 2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x + 1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

21 The functions f and g are such that $f(x) = x^2 - 2x$ $g(x) = x + 3$ The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

YOUR SOLUTION

02 Find an inverse

TRIGGER *"work out", "find", a missing value*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x + 2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x + 1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

21 The functions f and g are such that $f(x) = x^2 - 2x$ $g(x) = x + 3$ The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$ $h^{-1}(x) =$

YOUR SOLUTION

03 Use composite functions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x + 2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x + 1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

21 The functions f and g are such that $f(x) = x^2 - 2x$ $g(x) = x + 3$ The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

YOUR SOLUTION

04 Solve from function notation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 + 6, \quad g(x) = x - 10$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q17

Solve quadratic equation

$$f(x) = x^2 + 6, \quad g(x) = x - 10$$

(a)

$$g(3) = 3 - 10 = -7$$

Finish: $fg(3) = 55$, $x = 5$, excluded value 0, and $h^{-1}(x) = \frac{4}{2-x}$.

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 The functions f and g are defined as $f(x) = x^2 + 6$ $g(x) = x - 10$ (a) Find $fg(3)$ (2) (b) Solve the equation $fg(x) = f(x)$ Show clear algebraic working. (3) The function h is defined as $h(x) = 24x -$ (c) State the value of x that cannot be included in the domain of h (1) (d) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$ $h^{-1}(x) = (3)$

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = \frac{2x}{x-6}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q14

Evaluate fraction

$$f(x) = \frac{2x}{x-6}$$

(a)

$$f(10) = \frac{2(10)}{10-6} = \frac{20}{4} = 5$$

Finish: $f(10) = 5$, and $f^{-1}(x) = \frac{6x}{x-2}$.

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y , composite means work inside first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

14 The function f is defined as $f : x \mapsto 6x - x^2$ (a) Find $f(10)$ (1) (b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \dots$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Substitute into a function
"work out", "find", a missing value	Find an inverse
"work out", "show", "hence", a familiar exam phrase	Use composite functions
"work out", "show", "hence", a familiar exam phrase	Solve from function notation
"work out", "show", "hence", a familiar exam phrase	Evaluate fraction
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Linear Graphs ($y = mx + c$)

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

LINEAR GRAPHS ($Y = MX + C$) · CH3 · L4



ENG. ESLAM AHMED

The Map

This lesson collapses Linear Graphs ($y = mx + c$) into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Read gradient and intercept

TRIGGER *"gradient", "tangent", "rate of change"*

02 Find equation from two points

TRIGGER *"work out", "find", a missing value*

03 Use parallel/perpendicular gradients

TRIGGER *"gradient", "tangent", "rate of change"*

04 Sketch from equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 8 website questions, 8 checked solutions, 3 local PDFs read.

01 Read gradient and intercept

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5y + 2x = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2H Q18

The line is

$$5y + 2x = 7$$

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{5}{2}$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Find equation from two points

TRIGGER *"work out", "find", a missing value*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{0 - (-3)}{2 - 0} = \frac{3}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2023 P2H Q9

Solve equation

This is a reasoning step: write one clear sentence, then continue the calculation.

The gradient is

$$\frac{0 - (-3)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 3$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use parallel/perpendicular gradients

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5y + 2x = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2H Q18

The line is

$$5y + 2x = 7$$

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{5}{2}$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Sketch from equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Linear Graphs ($y = mx + c$) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m , intercept is c . Parallel lines have equal gradients.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"gradient", "tangent", "rate of change"	Read gradient and intercept
"work out", "find", a missing value	Find equation from two points
"gradient", "tangent", "rate of change"	Use parallel/perpendicular gradients
"work out", "show", "hence", a familiar exam phrase	Sketch from equation
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "find", a missing value	Find the gradient
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphs of Functions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

GRAPHS OF FUNCTIONS · CH3 · L5



ENG. ESLAM AHMED

The Map

This lesson collapses Graphs of Functions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 34 website questions, 34 checked solutions, 3 local PDFs read.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{x}{2} - 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q15

Solve quadratic equation

$$y = x^2 - \frac{x}{2} - 3$$

The completed table is

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

Finish: Missing values are 2, -1.5, -3, 0.

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

15 (a) Complete the table of values for $y \propto x^{-2}$

(b) On the grid, draw the graph of $y = x^2 - 2x + 3$ for values of x from -3 to 3.

YOUR SOLUTION

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x - 1)^2 - a$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x - 1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y -intercept $(0, 4 - a)$, x -intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

20 The curve C has equation $y = 4(x - 1)^2 - a$ where $a > 4$. Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a , (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. $y \times O$

YOUR SOLUTION

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

15 Here are six graphs. Graph A $y \propto x$ Graph B $y \propto x^2$ Graph C $y \propto \frac{1}{x}$ Graph D $y \propto \frac{1}{x^2}$
Graph E $y \propto x^3$ Graph F $y \propto x^4$ Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines.

Equation Graph $y = 2x$ $y = -1x$ $y = -5x$

YOUR SOLUTION

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x - 1)^2 - a$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x - 1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y -intercept $(0, 4 - a)$, x -intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

20 The curve C has equation $y = 4(x - 1)^2 - a$ where $a > 4$. Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a , (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. $y \times O$

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x^2 - 4x - 1 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q21

Read the graph

$$2x^2 - 4x - 1 = 0$$

Read the graph

$$x = -0.2, \quad x = 2.2$$

Finish: (a) $-0.2, 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6, 1.6$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.  (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$. Give your solutions correct to one decimal place. (2) (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$. Show your working clearly. Give your solutions correct to one decimal place. (3)

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = a \cos bx^\circ + c$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q26

The curve is

$$y = a \cos bx^\circ + c$$

Use trigonometry

3

Finish: $a = 2, b = 3, c = 1$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

26 Here is a sketch of the curve with equation $y = a \cos bx \text{ degrees} + c$ where $-90 < x < 450$

Find the value of a , the value of b and the value of c $a = b = c =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Complete a table of values
"draw", "plot", "construct", a dia- gram/grid	Plot the curve
"work out", "show", "hence", a familiar exam phrase	Read roots and intersections
"work out", "show", "hence", a familiar exam phrase	Transform the graph
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Estimating Gradients

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ESTIMATING GRADIENTS · CH₃ · L6



ENG. ESLAM AHMED

The Map

This lesson collapses Estimating Gradients into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw a tangent

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Use a large gradient triangle

TRIGGER *"gradient", "tangent", "rate of change"*

03 Estimate average gradient

TRIGGER *"gradient", "tangent", "rate of change"*

04 Interpret the rate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

06 Draw a tangent to the

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BANK EVIDENCE 3 website questions, 3 checked solutions, 3 local PDFs read.

01 Draw a tangent

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. (2) Use the graph to find the two possible values of k.

YOUR SOLUTION

02 Use a large gradient triangle

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. (2) Use the graph to find the two possible values of k.

YOUR SOLUTION

03 Estimate average gradient

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

04 Interpret the rate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. (2) Use the graph to find the two possible values of k .

YOUR SOLUTION

06 Draw a tangent to the

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6 , tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$. The point P on the curve has x coordinate 2. (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$. (2) (b) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$. Use the graph to find the two possible values of k .

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw a tangent
"gradient", "tangent", "rate of change"	Use a large gradient triangle
"gradient", "tangent", "rate of change"	Estimate average gradient
"work out", "show", "hence", a familiar exam phrase	Interpret the rate
"work out", "find", a missing value	Find the gradient
"draw", "plot", "construct", a dia- gram/grid	Draw a tangent to the

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Real-Life Graphs

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

REAL-LIFE GRAPHS · CH3 · L7

ENG. ESLAM AHMED



The Map

This lesson collapses Real-Life Graphs into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{x}{2} - 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q15

Solve quadratic equation

$$y = x^2 - \frac{x}{2} - 3$$

The completed table is

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

Finish: Missing values are 2, -1.5, -3, 0.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

15 (a) Complete the table of values for $y \propto x^{-2}$

(b) On the grid, draw the graph of $y = x^2 - 2x + 3$ for values of x from -3 to 3.

YOUR SOLUTION

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x - 1)^2 - a$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x - 1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y -intercept $(0, 4 - a)$, x -intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

20 The curve C has equation $y = 4(x - 1)^2 - a$ where $a > 4$. Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a , (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. $y \times O$

YOUR SOLUTION

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

15 Here are six graphs. Graph A Graph B Graph C Graph D Graph E Graph F Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines.

Equation Graph $y = 2x$ $y = -1x$ $y = -5x$

YOUR SOLUTION

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x - 1)^2 - a$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x - 1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y -intercept $(0, 4 - a)$, x -intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

20 The curve C has equation $y = 4(x - 1)^2 - a$ where $a > 4$. Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a , (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. $y \times O$

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x^2 - 4x - 1 = 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q21

Read the graph

$$2x^2 - 4x - 1 = 0$$

Read the graph

$$x = -0.2, \quad x = 2.2$$

Finish: (a) $-0.2, 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6, 1.6$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.  (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$. Give your solutions correct to one decimal place. (2) (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$. Show your working clearly. Give your solutions correct to one decimal place. (3)

YOUR SOLUTION

06 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

15 Here are six graphs. Graph A $y \propto x$ Graph B $y \propto x^2$ Graph C $y \propto \frac{1}{x}$ Graph D $y \propto \frac{1}{x^2}$
Graph E $y \propto x^3$ Graph F $y \propto x^4$ Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines.

Equation Graph $y = 2x$ $y = -1x$ $y = -5x$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Complete a table of values
"draw", "plot", "construct", a dia- gram/grid	Plot the curve
"work out", "show", "hence", a familiar exam phrase	Read roots and intersections
"work out", "show", "hence", a familiar exam phrase	Transform the graph
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	Solve equation

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphing Inequalities

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

GRAPHING INEQUALITIES · CH₃ · L8

ENG. ESLAM AHMED



The Map

This lesson collapses Graphing Inequalities into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw the boundary line

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Choose solid or dashed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Test one point

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Shade the correct region

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 The required region is

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 19 website questions, 19 checked solutions, 3 local PDFs read.

01 Draw the boundary line

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x \leq 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q10

The shaded region is left

$$x \leq 0$$

It is below the line

$$y \leq -x + 4$$

Finish: $x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

10 The shaded region in the diagram is bounded by three lines. The equation of one of the lines is given. $y = 1.3x - 2$ Write down the three inequalities that define the shaded region.

YOUR SOLUTION

02 Choose solid or dashed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2 < x \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q11

Solve inequality

$$2 < x \leq 4$$

The integer values are

3,4

Finish: (a) 3,4. (b) $y \geq x - 3$, $x + y \geq 5$, $y \leq 6$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

11 (a) Write down the integer values of x that satisfy the inequality $-2 < x < 4$ (2) The region R, shown shaded in the diagram, is bounded by three straight lines. Diagram NOT accurately drawn R O y $y = x - 3$ $x + y = 5$ $y = 6$ (b) Write down the three inequalities that define the region R. (2)

YOUR SOLUTION

03 Test one point

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -3x + 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q7

A straight line with gradient

$$y = -3x + 5$$

Use the inequalities

$$x < 6$$

Finish: (a) $y = -3x + 5$. (b) Shade $x < 6$, $y > 2$, $y \leq x + 1$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

7 (a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates (0, 5) (2) (b) Show, by shading on the grid, the region defined by all three of the inequalities $x < 6$ $y \geq 2$ $y < x + 1$ Label the region R O 2

4 6 8 10 10 8 6 4 2 x y (3)

YOUR SOLUTION

04 Shade the correct region

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x \leq 0$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q10

The shaded region is left

$$x \leq 0$$

It is below the line

$$y \leq -x + 4$$

Finish: $x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 - 4x > 17$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q8

Read the graph

$$9 - 4x > 17$$

Read the graph

$$y \geq 2$$

Finish: (a) $x < -2$. (b) $y \geq 2, x \leq 6, y \leq x$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 (a) Solve $9 - 4x > 17$ (2) O 1 2 3 4 5 6 7 8 1 2 3 4 5 6 7 8 9 10 y x R (b) Write down the three inequalities that represent the shaded region R (3)

YOUR SOLUTION

06 The required region is

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 1, \quad x = 2, \quad x + y = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q2

Draw the boundary lines

$$y = 1, \quad x = 2, \quad x + y = 7$$

The required region is

$$y \geq 1$$

Finish: Shade $y \geq 1, x \geq 2, x + y \leq 7$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw the boundary line
"work out", "show", "hence", a familiar exam phrase	Choose solid or dashed
"work out", "show", "hence", a familiar exam phrase	Test one point
"work out", "show", "hence", a familiar exam phrase	Shade the correct region
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	The required region is

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations of Graphs

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

TRANSFORMATIONS OF GRAPHS · CH₃ · L₉

ENG. ESLAM AHMED



The Map

This lesson collapses Transformations of Graphs into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Shift up or down

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Shift left or right

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Stretch vertically

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Reflect or scale horizontally

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 38 website questions, 38 checked solutions, 3 local PDFs read.

01 Shift up or down

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. $y \times 360$
 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b . $a = b = (2)$
 Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are
 (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation
 (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

02 Shift left or right

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. $y \times 360$
 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b . $a = b = (2)$
 Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are
 (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation
 (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

03 Stretch vertically

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = f(x + 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q21

The maximum point of is

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$y = f(x + 4)$$

Finish: $(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Reflect or scale horizontally

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = f(x + 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q21

The maximum point of is

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$y = f(x + 4)$$

Finish: $(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$. The point $P(4, 6)$ is a maximum point on the curve. The coordinates of the other point where the curve crosses the y -axis are $(0, 4)$. The curve crosses the x -axis at $(2, 0)$ and $(6, 0)$. (a) Write down the coordinates of the maximum point on the curve with equation (i) $y = f(x + 4)$ (ii) $y = f(2x)$ (2) The equation of a curve C is $y = x^2 + 3x + 4$. The curve C is transformed to curve S under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$. (b) Find an equation of curve S . You do not need to simplify the equation. (2)

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. $y \times 360$
 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b . $a = b = (2)$
 Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are
 (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation
 (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Shift up or down
"work out", "show", "hence", a familiar exam phrase	Shift left or right
"work out", "show", "hence", a familiar exam phrase	Stretch vertically
"work out", "show", "hence", a familiar exam phrase	Reflect or scale horizontally
"work out", "show", "hence", a familiar exam phrase	Read the graph

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Differentiation

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

DIFFERENTIATION · CH₃ · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Differentiation into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Differentiate powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find gradient at a point

TRIGGER *"work out", "find", a missing value*

03 Find stationary points

TRIGGER *"work out", "find", a missing value*

04 Use derivative in a context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Differentiate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 40 website questions, 40 checked solutions, 3 local PDFs read.

01 Differentiate powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = (2x + 5)(x + 1)(3 - x)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q15

The volume is

$$V = (2x + 5)(x + 1)(3 - x)$$

Expand brackets

$$(2x + 5)(x + 1) = 2x^2 + 7x + 5$$

Finish: $x = 1.47$ to 3 significant figures.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

15 (2x + 5) cm (x + 1) cm (3 - x) cm Diagram NOT accurately drawn The diagram shows a cuboid of volume $V \text{ cm}^3$ (a) Show that $V = 15 + 16x - x^2 - 2x^3$ (3) There is a value of x for which the volume of the cuboid is a maximum. (b) Find this value of x . Show your working clearly. Give your answer correct to 3 significant figures. $x = (5)$

YOUR SOLUTION

02 Find gradient at a point

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = ax^3 + bx^2 - 12x + 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q24

Calculate value

$$y = ax^3 + bx^2 - 12x + 6$$

Point lies on the curve

$$-6 = 8a + 4b - 24 + 6$$

Finish: $y = 33$

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

24 The curve C has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants. The point A with coordinates (2, -6) lies on C. The gradient of the curve at A is 16. Find the y coordinate of the point on the curve whose x coordinate is 3. Show clear algebraic working.

y = O A S 100 A S

YOUR SOLUTION

03 Find stationary points

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^3 - 8x^2 - 12x + 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1HR Q19

Differentiate

$$y = x^3 - 8x^2 - 12x + 5$$

Stationary points have

$$3x^2 - 16x - 12 = 0$$

Finish: $A = (6, -139)$.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

19 A curve C has equation $y = x^3 - 2x^2 + 12x - 5$. Curve C has exactly two stationary points, one at point A and one at point B such that x coordinate of point A > x coordinate of point B. Find the coordinates of point A. Show clear algebraic working. (,)

YOUR SOLUTION

04 Use derivative in a context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q17

Volume of the solid is

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

Finish: The maximum value of y is 320.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Differentiate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q17

Volume of the solid is

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

Finish: The maximum value of y is 320.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

17 A solid, S, is made from a hemisphere and a cylinder. The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point. x cm $(20 - 4x)$ cm Diagram NOT accurately drawn The radius of the cylinder and the radius of the hemisphere are both x cm. The height of the cylinder is $(20 - 4x)$ cm. The volume of S is V cm³ where $V = \frac{1}{3}\pi y$ Find the maximum value of y . Show clear algebraic working.

YOUR SOLUTION

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{p}{x}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q22

Find the gradient

$$y = x^2 - \frac{p}{x}$$

Find the gradient

$$2(-3) + \frac{p}{(-3)^2} = 0$$

Finish: $p = 54$, $k = 12$

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

22 The diagram shows a sketch of part of the curve with equation $y = x^2 - px$ where p is a positive constant. Diagram NOT accurately drawn

y x O

For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch. For the curve where the x coordinate of T is -3 (a) find the value of p $p = (4)$ The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - 16x$ (b) Find the value of k $k = (3)$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Differentiate powers
"work out", "find", a missing value	Find gradient at a point
"work out", "find", a missing value	Find stationary points
"work out", "show", "hence", a familiar exam phrase	Use derivative in a context
"work out", "show", "hence", a familiar exam phrase	Differentiate
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Standard & Compound Units

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

STANDARD & COMPOUND UNITS · CH4 · L1



ENG. ESLAM AHMED

The Map

This lesson collapses Standard & Compound Units into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Pressure: force over area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Density: mass and volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Speed: distance over time

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Volume unit conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 39 website questions, 39 checked solutions, 4 local PDFs read.

01 Pressure: force over area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q13

Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The cube rests on one

$$w^2 = 400$$

Finish: 8000 cm³.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A cube rests on one face. The force on the table is 72 N and the pressure is 0.18 N/cm^2 . Find the volume of the cube.

YOUR SOLUTION

02 Density: mass and volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{mass} = \text{density} \times \text{volume}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q9

Calculate volume

$$\text{mass} = \text{density} \times \text{volume}$$

Finish: 2895 g.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A metal cuboid has mass 864 g and volume 320 cm³. Work out its density, then decide whether it could be aluminium with density 2.7 g/cm³.

YOUR SOLUTION

03 Speed: distance over time

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q5

The time is

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

Average speed

$$2470 \div 3.6 = 686.111 \dots$$

Finish: 686 km/h.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Volume unit conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1 \text{ m} = 100 \text{ cm}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q2

Calculate value

$$1 \text{ m} = 100 \text{ cm}$$

Calculate value

$$1 \text{ m}^3 = 100^3 \text{ cm}^3 = 1,000,000 \text{ cm}^3$$

Finish: 32,400,000 cm³.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Pressure: force over area
"work out", "show", "hence", a familiar exam phrase	Density: mass and volume
"work out", "show", "hence", a familiar exam phrase	Speed: distance over time
"work out", "show", "hence", a familiar exam phrase	Volume unit conversion

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Angles in Polygons & Parallel Lines

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ANGLES IN POLYGONS & PARALLEL LINES · CH4 · L2

ENG. ESLAM AHMED



The Map

This lesson collapses Angles in Polygons & Parallel Lines into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 20 website questions, 20 checked solutions, 4 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$30 + (4x + 10) + (x + 20) = 180$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q4

Angles in a triangle add

$$30 + (4x + 10) + (x + 20) = 180$$

Finish: $x = 24$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

4 The diagram shows a triangle. Diagram NOT accurately drawn 30 degrees ($4x + 10$) degrees ($x + 20$) degrees Work out the value of x . $x =$

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$30 + (4x + 10) + (x + 20) = 180$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q4

Angles in a triangle add

$$30 + (4x + 10) + (x + 20) = 180$$

Finish: $x = 24$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(9x + 27) + (2x - 23) = 180$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 P1H Q12

The interior angle and exterior

$$(9x + 27) + (2x - 23) = 180$$

The exterior angle is

$$2x - 23 = 2(16) - 23 = 9^\circ$$

Finish: $n = 40$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

177, 175, 173, 171, 169, ...

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q23

The interior angles form an

177, 175, 173, 171, 169, ...

Calculate value

$$a = 177, \quad d = -2$$

Finish: $n = 18$

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{(5-2)180}{5} = 108^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q8

A regular pentagon has interior

$$\frac{(5-2)180}{5} = 108^\circ$$

A regular hexagon has interior

$$\frac{(6-2)180}{6} = 120^\circ$$

Finish: 96° .

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: 360° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 The diagram shows a regular pentagon, ABCDE, a regular hexagon, CFGHID, and a quadrilateral, EDIJ. A B F G H C D I x J E Diagram NOT accurately drawn AEJ and HIJ are straight lines. Work out the size of the angle marked x. Show your working clearly. degrees

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

177, 175, 173, 171, 169, ...

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q23

The interior angles form an

177, 175, 173, 171, 169, ...

Calculate value

$$a = 177, \quad d = -2$$

Finish: $n = 18$

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "find", a missing value	Find the gradient
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Bearings, Scale Drawing & Constructions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

BEARINGS, SCALE DRAWING & CONSTRUCTIONS · CH4 · L3

ENG. ESLAM AHMED



The Map

This lesson collapses Bearings, Scale Drawing & Constructions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw North first

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Measure clockwise bearing

TRIGGER *"bearing", "north", "scale drawing"*

03 Use scale before real distance

TRIGGER *"bearing", "north", "scale drawing"*

04 Construct the locus

TRIGGER *"draw", "plot", "construct", a diagram/grid*

05 Use bearings

TRIGGER *"bearing", "north", "scale drawing"*

06 Construction answer

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Draw North first

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q16

Use bearings

This is a reasoning step: write one clear sentence, then continue the calculation.

Resolve

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

Finish: 140° .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

16 The diagram shows the positions of three ships, A, B and C. A 150 m 275 m North North 120 degrees B C Diagram NOT accurately drawn Ship B is due north of ship A. The bearing of ship C from ship A is 120 degrees Calculate the bearing of ship C from ship B. Give your answer correct to the nearest degree. degrees

YOUR SOLUTION

02 Measure clockwise bearing

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

03 Use scale before real distance

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1 cm represents 200 m

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q5

The distance from to is

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale is

1 cm represents 200 m

Finish: Draw C 3.5 cm from A on a bearing of 110° , and the scale is 1 : 20000.

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

5 The scale diagram shows the position on a map of a house, A A North Scale: 1 cm represents 200 metres House C is on a bearing of 110 degrees from A The distance from A to C is 700 m (a) Mark the position of C on the diagram with a cross (times) Label your cross C (3) (b) Write the scale of the map in the form 1 : n 1 : (1)

04 Construct the locus

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

05 Use bearings

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q16

Use bearings

This is a reasoning step: write one clear sentence, then continue the calculation.

Resolve

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

Finish: 140° .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

16 The diagram shows the positions of three ships, A, B and C. A 150 m 275 m North North 120 degrees B C Diagram NOT accurately drawn Ship B is due north of ship A. The bearing of ship C from ship A is 120 degrees Calculate the bearing of ship C from ship B. Give your answer correct to the nearest degree. degrees

YOUR SOLUTION

06 Construction answer

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw North first
"bearing", "north", "scale drawing"	Measure clockwise bearing
"bearing", "north", "scale drawing"	Use scale before real distance
"draw", "plot", "construct", a dia- gram/grid	Construct the locus
"bearing", "north", "scale drawing"	Use bearings
"draw", "plot", "construct", a dia- gram/grid	Construction answer

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circle Theorems

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CIRCLE THEOREMS · CH4 · L4

ENG. ESLAM AHMED



The Map

This lesson collapses Circle Theorems into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Spot the theorem family

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use tangent-radius or alternate segment

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use cyclic quadrilateral angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Write the reason with the answer

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use circle theorem

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 31 website questions, 31 checked solutions, 5 local PDFs read.

01 Spot the theorem family

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q14

Angle and central angle stand

This is a reasoning step: write one clear sentence, then continue the calculation.

The angle at the centre

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Finish: 52° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use tangent-radius or alternate segment

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle BFD = 39^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q14

Use trigonometry

$$\angle BFD = 39^\circ$$

Given

$$\angle EFD = 18^\circ$$

Finish: 123° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use cyclic quadrilateral angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2021 P2H Q18

Use trigonometry

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

Opposite angles in a cyclic

$$\angle ADC = 180^\circ - 98^\circ = 82^\circ$$

Finish: 32.8 cm.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

04 Write the reason with the answer

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle RQP = 90^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q14

Use trigonometry

$$\angle RQP = 90^\circ$$

Reason the angle in a

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\angle RQP = 90^\circ$, and $\angle PSQ = 22^\circ$.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q14

Angle and central angle stand

This is a reasoning step: write one clear sentence, then continue the calculation.

The angle at the centre

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Finish: 52° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Use circle theorem

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m = \frac{124}{2} = 62$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q13

Use trigonometry

$$m = \frac{124}{2} = 62$$

Use circle theorem

$$n = 180 - 62 = 118$$

Finish: $m = 62$, $n = 118$, and $\angle QPO = 62^\circ$.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Spot the theorem family
"work out", "show", "hence", a familiar exam phrase	Use tangent-radius or alternate segment
"work out", "show", "hence", a familiar exam phrase	Use cyclic quadrilateral angles
"work out", "show", "hence", a familiar exam phrase	Write the reason with the answer
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use circle theorem
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Area & Perimeter

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

AREA & PERIMETER · CH4 · L5

ENG. ESLAM AHMED



The Map

This lesson collapses Area & Perimeter into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Split the compound shape

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Build an equation from area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use perimeter after missing sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check units squared or linear

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 32 website questions, 32 checked solutions, 4 local PDFs read.

01 Split the compound shape

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$12 \times 9 = 108 \text{ cm}^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q3

Split the shape into two

This is a reasoning step: write one clear sentence, then continue the calculation.

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

Finish: 7 cm.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Build an equation from area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = \frac{1}{2}\pi r^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q5

The garden is a semicircle

$$\text{area} = \frac{1}{2}\pi r^2$$

Calculate area

$$12 \times 6 = 72 \text{ m}^2$$

Finish: No, he does not have enough grass seed.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 The diagram shows Yuen's garden. 7.2 m Diagram NOT accurately drawn The garden is in the shape of a semicircle of radius 7.2 m. Yuen is going to cover his garden with grass seed. Yuen has 12 boxes of grass seed. Each box of grass seed contains enough seed to cover 6 m² of the garden. Has Yuen enough grass seed for his garden? Show your working clearly.

YOUR SOLUTION

03 Use perimeter after missing sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$12 \times 9 = 108 \text{ cm}^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q3

Split the shape into two

This is a reasoning step: write one clear sentence, then continue the calculation.

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

Finish: 7 cm.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Check units squared or linear

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AD = 17 \text{ cm}, \quad CD = 15 \text{ cm}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q11

Use trigonometry

$$AD = 17 \text{ cm}, \quad CD = 15 \text{ cm}$$

Use Pythagoras to find

$$AC^2 = 17^2 - 15^2 = 289 - 225 = 64$$

Finish: 44.6 cm.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

11 The diagram shows a shaded shape ABCD made from a semicircle ABC and a right-angled triangle ACD. Diagram NOT accurately drawn

17 cm 15 cm

A B C D

AC is the diameter of the semicircle ABC. Work out the perimeter of the shaded shape. Give your answer correct to 3 significant figures. cm

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$PQ = p$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q20

Let

$$PQ = p$$

Given

$$AB = 1.5PQ$$

Finish: $PQ = 4.8 \text{ cm}$

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

20 A B F P Q C D R E Diagram NOT accurately drawn T The diagram shows a shaded region T formed by removing an equilateral triangle PQR from a regular hexagon ABCDEF. The points P and Q lie on AB such that $AB = 1.5$ times PQ Given that the area of region T is 72.3 cm^2 work out the length of PQ. cm

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Split the compound shape
"work out", "show", "hence", a familiar exam phrase	Build an equation from area
"work out", "show", "hence", a familiar exam phrase	Use perimeter after missing sides
"work out", "show", "hence", a familiar exam phrase	Check units squared or linear
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circles, Arcs & Sectors

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CIRCLES, ARCS & SECTORS · CH4 · L6

ENG. ESLAM AHMED



The Map

This lesson collapses Circles, Arcs & Sectors into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use fraction of the circle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find sector area

TRIGGER *"work out", "find", a missing value*

03 Find arc length

TRIGGER *"work out", "find", a missing value*

04 Combine sector with triangle/rectangle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 41 website questions, 41 checked solutions, 4 local PDFs read.

01 Use fraction of the circle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 + 2 = 9 \text{ m}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q8

The inner radius is m

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$7 + 2 = 9 \text{ m}$$

Finish: 54.3 m.

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

8 The region, shown shaded in the diagram, is a path. Diagram NOT accurately drawn
2 m O 7 m The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines. The inner semicircle has a radius of 7 metres. The path has a width of 2 metres. Work out the perimeter of the path. Give your answer correct to one decimal place. m

YOUR SOLUTION

02 Find sector area

TRIGGER *"work out", "find", a missing value*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 + 2 = 9 \text{ m}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q8

The inner radius is m

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$7 + 2 = 9 \text{ m}$$

Finish: 54.3 m.

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

8 The region, shown shaded in the diagram, is a path. Diagram NOT accurately drawn
 2 m O 7 m The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines. The inner semicircle has a radius of 7 metres. The path has a width of 2 metres. Work out the perimeter of the path. Give your answer correct to one decimal place. m

YOUR SOLUTION

03 Find arc length

TRIGGER *"work out", "find", a missing value*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5 = \frac{55}{360} \times 2\pi r$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q16

Use trigonometry

$$5 = \frac{55}{360} \times 2\pi r$$

Area of the circle

$$\pi r^2 = \pi(5.208\dots)^2 = 85.233\dots$$

Finish: 85.2 cm².

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

16 The diagram shows a circle with centre O. Diagram NOT accurately drawn. A sector OBC has a central angle of 55°. B and C are points on the circle so that the length of the arc BC is 5 cm. Given that $\angle AOC = 55^\circ$, work out the area of the circle. Give your answer correct to one decimal place. cm^2

YOUR SOLUTION

04 Combine sector with triangle/rectangle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6 + x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q22

The radius of sector is

$$6 + x$$

Use trigonometry

$$\frac{50}{360} \times 2\pi(6 + x)$$

Finish: 5.89

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm. A 50 degrees D x cm O B C Diagram NOT accurately drawn 6 cm A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm Angle BOC = 50 degrees Given that the perimeter of sector OBC = 2 times the perimeter of triangle OAD find the value of x . Give your answer correct to 3 significant figures. $x =$

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6 + x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q22

The radius of sector is

$$6 + x$$

Use trigonometry

$$\frac{50}{360} \times 2\pi(6 + x)$$

Finish: 5.89

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm. A 50 degrees D x cm O B C Diagram NOT accurately drawn 6 cm A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm Angle BOC = 50 degrees Given that the perimeter of sector OBC = 2 times the perimeter of triangle OAD find the value of x . Give your answer correct to 3 significant figures. $x =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use fraction of the circle
"work out", "find", a missing value	Find sector area
"work out", "find", a missing value	Find arc length
"work out", "show", "hence", a familiar exam phrase	Combine sector with triangle/rectangle
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Volume & Surface Area

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

VOLUME & SURFACE AREA · CH4 · L7

ENG. ESLAM AHMED



The Map

This lesson collapses Volume & Surface Area into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the solid

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use prism volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Unfold surface area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Reverse from volume or area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 39 website questions, 39 checked solutions, 4 local PDFs read.

01 Name the solid

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q17

The cross section is a

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Calculate area

$$\frac{37 - 28}{2} = 4.5$$

Finish: 3844 cm², and 39.3°.

FORMULA BANK

Prism volume = area of cross-section × length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

17 The diagram shows a solid prism ABCDEFGH. A 28 cm 24 cm 20 cm D E F G H B C Diagram NOT accurately drawn 37 cm The trapezium ABCD, in which AD is parallel to BC, is a cross section of the prism. The base ADEH of the prism is a horizontal plane. ADEH and BCFG are rectangles. The midpoint of BC is vertically above the midpoint of AD so that BA = CD. AD = 37 cm GF = 28 cm DE = 24 cm The perpendicular distance between edges AD and BC is 20 cm. (a) Work out the total surface area of the prism. cm² (4) (b) Calculate ...

YOUR SOLUTION

02 Use prism volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{2}(7 + 12)h = 9.5h$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q13

Area of the trapezium cross

$$\frac{1}{2}(7 + 12)h = 9.5h$$

Volume of the prism

$$9.5h \times 10 = 608$$

Finish: $h = 6.4$ cm.

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

13 Diagram NOT accurately drawn A B D C 7 cm E 10 cm 12 cm h cm F G H The diagram shows a prism ABCDEFGH in which ABCD is a trapezium with BC parallel to AD and CDEF is a rectangle. BC = 7 cm AD = 12 cm DE = 10 cm The height of trapezium ABCD is h cm The volume of the prism is 608 cm³ Work out the value of h. h =

YOUR SOLUTION

03 Unfold surface area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{3}\pi r^2 h = 1600$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q26

Let the radius of the

This is a reasoning step: write one clear sentence, then continue the calculation.

The volume of the cone

$$\frac{1}{3}\pi r^2 h = 1600$$

Finish: 107°

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

26 Here is a sector, AOB, of a circle with centre O and angle AOB = x degrees. Diagram NOT accurately drawn. The sector can form the curved surface of a cone by joining OA to OB. Diagram NOT accurately drawn. The height of the cone is 25 cm. The volume of the cone is 1600 cm³. Work out the value of x . Give your answer correct to the nearest whole number. ' x =

YOUR SOLUTION

04 Reverse from volume or area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{3}\pi r^2 h = 1600$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q26

Let the radius of the

This is a reasoning step: write one clear sentence, then continue the calculation.

The volume of the cone

$$\frac{1}{3}\pi r^2 h = 1600$$

Finish: 107°

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

26 Here is a sector, AOB, of a circle with centre O and angle AOB = x degrees. Diagram NOT accurately drawn. The sector can form the curved surface of a cone by joining OA to OB. Diagram NOT accurately drawn. The height of the cone is 25 cm. The volume of the cone is 1600 cm³. Work out the value of x . Give your answer correct to the nearest whole number. ' x =

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q17

The cross section is a

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Calculate area

$$\frac{37 - 28}{2} = 4.5$$

Finish: 3844 cm², and 39.3°.

FORMULA BANK

Prism volume = area of cross-section × length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

17 The diagram shows a solid prism ABCDEFGH. A 28 cm 24 cm 20 cm D E F G H B C Diagram NOT accurately drawn 37 cm The trapezium ABCD, in which AD is parallel to BC, is a cross section of the prism. The base ADEH of the prism is a horizontal plane. ADEH and BCFG are rectangles. The midpoint of BC is vertically above the midpoint of AD so that BA = CD. AD = 37 cm GF = 28 cm DE = 24 cm The perpendicular distance between edges AD and BC is 20 cm. (a) Work out the total surface area of the prism. cm² (4) (b) Calculate ...

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the solid
"work out", "show", "hence", a familiar exam phrase	Use prism volume
"work out", "show", "hence", a familiar exam phrase	Unfold surface area
"work out", "show", "hence", a familiar exam phrase	Reverse from volume or area
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Congruence, Similarity & Geometrical Proof

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CONGRUENCE, SIMILARITY & GEOMETRICAL PROOF · CH4 · L8



ENG. ESLAM AHMED

The Map

This lesson collapses Congruence, Similarity & Geometrical Proof into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

05 Evaluate fraction

TRIGGER *"similar", "scale factor", "area/volume"*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 11 website questions, 11 checked solutions, 4 local PDFs read.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{AC}{CE} = \frac{BC}{CD}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Finish: $x = \frac{9}{8}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 Diagram NOT accurately drawn A (2x + 9) cm 2x cm 1.2 cm 6 cm D E B C ACE and BCD are straight lines. AB is parallel to DE Work out the value of x x =

YOUR SOLUTION

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$CD = 1.5BE$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Find the gradient

$$CD = 1.5BE$$

Finish: (a) $ED = 5$ cm, (b) $x = 3.75$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 10 cm Diagram NOT accurately drawn E D C A B In the diagram, ABC and AED are straight lines. BE is parallel to CD AE = 10 cm and CD = 1.5 times BE (a) Work out the length of ED cm (2) AB = $(2x + 5)$ cm and BC = $(3x - 5)$ cm (b) Work out the value of x $x = (2)$

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{AC}{CE} = \frac{BC}{CD}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Finish: $x = \frac{9}{8}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

12 Diagram NOT accurately drawn A (2x + 9) cm 2x cm 1.2 cm 6 cm D E B C ACE and BCD are straight lines. AB is parallel to DE Work out the value of x x =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find the length scale factor
"similar", "scale factor", "area/volume"	Square it for area
"similar", "scale factor", "area/volume"	Cube it for volume
"similar", "scale factor", "area/volume"	Reverse from area or volume
"similar", "scale factor", "area/volume"	Evaluate fraction
"work out", "find", a missing value	Find the gradient
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Area & Volume of Similar Shapes

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

AREA & VOLUME OF SIMILAR SHAPES · CH4 · L9

ENG. ESLAM AHMED



The Map

This lesson collapses Area & Volume of Similar Shapes into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

05 Split the ratio

TRIGGER *"similar", "scale factor", "area/volume"*

BANK EVIDENCE 36 website questions, 36 checked solutions, 3 local PDFs read.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$15 : 45 = 1 : 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2023 P1H Q23

The small and large square

This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are

$$15 : 45 = 1 : 3$$

Finish: 5090 cm²

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram. Diagram NOT accurately drawn

15 cm 39 cm 39 cm 45 cm

The height of the small pyramid is 15 cm. The height of the large pyramid is 45 cm. The square base of the large pyramid has side length 39 cm. Work out the total surface area of the frustum. Give your answer correct to the nearest whole number. cm²

YOUR SOLUTION

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$15 : 45 = 1 : 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2023 P1H Q23

The small and large square

This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are

$$15 : 45 = 1 : 3$$

Finish: 5090 cm²

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram. Diagram NOT accurately drawn
15 cm 39 cm 39 cm 45 cm The height of the small pyramid is 15 cm. The height of the large pyramid is 45 cm. The square base of the large pyramid has side length 39 cm. Work out the total surface area of the frustum. Give your answer correct to the nearest whole number. cm²

YOUR SOLUTION

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{h}{2h} = \frac{1}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q20

The two cones are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

The large cone has height

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $r = \frac{7h}{16}$

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

20 The diagram shows a frustum of a cone and a sphere. The frustum is made by removing a small cone from a large cone. The cones are similar. h cm $2h$ cm r cm r cm
Diagram NOT accurately drawn The height of the small cone is h cm. The height of the large cone is $2h$ cm. The radius of the base of the large cone is r cm. The radius of the sphere is r cm. Given that the volume of the frustum is equal to the volume of the sphere, find an expression for r in terms of h . Give your expression in its simplest form. $r =$

YOUR SOLUTION

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{h}{2h} = \frac{1}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q20

The two cones are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

The large cone has height

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $r = \frac{7h}{16}$

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Split the ratio

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $9 : 13$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q20

The two vases are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are in the

$9 : 13$

Finish: 583.2 cm^2

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

20 The diagram shows two similar vases, A and B. Diagram NOT accurately drawn
 9 cm 13 cm A B The height of vase A is 9 cm and the height of vase B is 13 cm. Given
 that surface area of vase A + surface area of vase B = 1800 cm² calculate the surface
 area of vase A. cm²

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find the length scale factor
"similar", "scale factor", "area/volume"	Square it for area
"similar", "scale factor", "area/volume"	Cube it for volume
"similar", "scale factor", "area/volume"	Reverse from area or volume
"similar", "scale factor", "area/volume"	Split the ratio

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Right-Angled Triangles - Pythagoras & Trigonometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RIGHT-ANGLED TRIANGLES - PYTHAGORAS & TRIGONOMETRY · CH4 · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Right-Angled Triangles - Pythagoras & Trigonometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Angle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 32 website questions, 32 checked solutions, 3 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$EF = 12 \cos 40^\circ = 9.192 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q15

Use trigonometry

$$EF = 12 \cos 40^\circ = 9.192 \dots$$

Use trigonometry

$$\sin 28^\circ = \frac{EF}{EG}$$

Finish: 19.6 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$BD = 16, \quad \angle DAB = 65^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q10

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Use trigonometry

$$BD = 16, \quad \angle DAB = 65^\circ$$

Finish: 50.2 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

06 Angle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AB^2 = 21^2 - 15^2 = 216$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM1H Q11

Use trigonometry

$$AB^2 = 21^2 - 15^2 = 216$$

Angle

$$\cos ACB = \frac{15}{21}$$

Finish: 13.5° .

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Angle

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sine, Cosine Rule & Area of Triangles

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SINE, COSINE RULE & AREA OF TRIANGLES · CH4 · L11

ENG. ESLAM AHMED



The Map

This lesson collapses Sine, Cosine Rule & Area of Triangles into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use cosine rule

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 24 website questions, 24 checked solutions, 3 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$BC = 13 \sin 40^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q21

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

Find the gradient

$$BC = 13 \sin 40^\circ$$

Finish: 12.4 cm

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{BC}{2} = 14$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q9

Use cosine rule

$$\frac{BC}{2} = 14$$

Finish: 147 cm².

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

06 Use cosine rule

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{BC}{2} = 14$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q9

Use cosine rule

$$\frac{BC}{2} = 14$$

Finish: 147 cm².

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use cosine rule

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

3D Pythagoras & Trigonometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

3D PYTHAGORAS & TRIGONOMETRY · CH4 · L12

ENG. ESLAM AHMED



The Map

This lesson collapses 3D Pythagoras & Trigonometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw the hidden right triangle

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Find the base diagonal first

TRIGGER *"work out", "find", a missing value*

03 Use 3D trig after Pythagoras

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Give the final angle/length

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the midpoint

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Draw the hidden right triangle

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q22

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The line from to the

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

Finish: 74.7°

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

YOUR SOLUTION

02 Find the base diagonal first

TRIGGER *"work out", "find", a missing value*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q22

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The line from to the

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

Finish: 74.7°

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use 3D trig after Pythagoras

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q19

The projection of onto the

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

Use trigonometry

$$\tan \theta = \frac{6}{\sqrt{106}}$$

Finish: 30.2°.

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

YOUR SOLUTION

04 Give the final angle/length

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AB = 15, \quad BC = 4, \quad CF = 9$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q22

Find the gradient

$$AB = 15, \quad BC = 4, \quad CF = 9$$

Use trigonometry

$$PR = 42$$

Finish: $BE = 17.9 \text{ cm}$, $\angle UMR = 49.1^\circ$

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q19

The projection of onto the

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

Use trigonometry

$$\tan \theta = \frac{6}{\sqrt{106}}$$

Finish: 30.2°.

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

YOUR SOLUTION

06 Find the midpoint

TRIGGER *"work out", "find", a missing value*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AE = x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q21

Let

$$AE = x$$

Use trigonometry

$$AB = BD = DE = AE = x$$

Finish: $T = \frac{\sqrt{68} + \sqrt{51}}{17}$

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw the hidden right triangle
"work out", "find", a missing value	Find the base diagonal first
"work out", "show", "hence", a familiar exam phrase	Use 3D trig after Pythagoras
"work out", "show", "hence", a familiar exam phrase	Give the final angle/length
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "find", a missing value	Find the midpoint

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Vectors

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

VECTORS · CH5 · L1

ENG. ESLAM AHMED



The Map

This lesson collapses Vectors into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Follow the route

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use equal parallel vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use midpoint or ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Prove collinearity

TRIGGER *"show that", "prove", "give a reason"*

05 Use vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 41 website questions, 41 checked solutions, 4 local PDFs read.

01 Follow the route

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q14

Use vectors

$$\overrightarrow{OA} = \mathbf{a}, \quad \overrightarrow{OB} = \mathbf{b}$$

(a)(i)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$$

Finish: (i) $\mathbf{b} - \mathbf{a}$, (ii) $4\mathbf{b} - 2\mathbf{a}$, (iii) $2\mathbf{b} - 3\mathbf{a}$, (b) 90 cm^2 .

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

14 ABCDEF and GHIJKL are regular hexagons each with centre O. Diagram NOT accurately drawn G L I H K J E D A B F C O a b GHIJKL is an enlargement of ABCDEF, with centre O and scale factor 2 OA to = a OB to = b (a) Write the following vectors, in terms of a and b. Simplify your answers. (i) AB to (1) (ii) KI to (2) (iii) LD to (2) The triangle OAB has an area of 5 cm² (b) Calculate the area of the shaded region. cm² (3)

YOUR SOLUTION

02 Use equal parallel vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q25

Let be the origin

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

Read the graph

$$B = \mathbf{a}, \quad D = \mathbf{b}, \quad C = \mathbf{a} + \mathbf{b}$$

Finish: $\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

25 ABCD is a parallelogram and ADM is a straight line. Diagram NOT accurately drawn A D K M B L C AB to = a BC to = b DM to = 1 2 b K is the point on AB such that $AK : AB = \lambda : 1$ L is the point on CD such that $CL : CD = \mu : 1$ KLM is a straight line. Given that $\lambda : \mu = 1 : 2$ use a vector method to find the value of λ and the value of μ $\lambda = \mu =$ O A O A S 100 A S

YOUR SOLUTION

03 Use midpoint or ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q23

Split the ratio

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

Finish: $AP : PM = 3 : 1$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

23 Diagram NOT accurately drawn 4a 2b A B C M P ABC is a triangle. The midpoint of BC is M. P is a point on AM. AB to = 4a AC to = 2b AP to = 3 2 a + 3 4 b Find the ratio AP : PM

YOUR SOLUTION

04 Prove collinearity

TRIGGER *"show that", "prove", "give a reason"*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q26

Read the graph

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

Read the graph

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Finish: $\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

26 The diagram shows trapezium OACB. Diagram NOT accurately drawn A B C O
4b 6b 3a OA to = 3a OB to = 6b AC to = 4b N is the point on OC such that ANB is a straight line. Find ON to as a simplified expression in terms of a and b.

YOUR SOLUTION

05 Use vectors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q23

Use vectors

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

Find the midpoint

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

Finish: $OP : PM = 4 : 1$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Vectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q26

Read the graph

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

Read the graph

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Finish: $\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$

FORMULA BANK

Route vectors by arrows. Equal parallel arrows have equal vectors. Section ratio controls the fractions on the same line.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

26 The diagram shows trapezium OACB. Diagram NOT accurately drawn

A B C O

4b 6b 3a OA to = 3a OB to = 6b AC to = 4b

N is the point on OC such that ANB is a straight line. Find ON to as a simplified expression in terms of a and b.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Follow the route
"work out", "show", "hence", a familiar exam phrase	Use equal parallel vectors
"work out", "show", "hence", a familiar exam phrase	Use midpoint or ratio
"show that", "prove", "give a reason"	Prove collinearity
"work out", "show", "hence", a familiar exam phrase	Use vectors
"work out", "show", "hence", a familiar exam phrase	Read the graph
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

TRANSFORMATIONS · CH5 · L15

ENG. ESLAM AHMED



The Map

This lesson collapses Transformations into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Identify the transformation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use centre and scale factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Write the vector translation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Describe a combined movement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use Transformations tag correct

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 10 website questions, 10 checked solutions, 7 local PDFs read.

01 Identify the transformation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q8

Use

$$\text{new point} = (1,2) + \frac{1}{2}(\text{old point} - (1,2))$$

The vertices

$$(7,10), (15,10), (15,4), (11,4)$$

Finish: draw the enlarged shape with vertices $(4,6), (8,6), (8,3), (6,3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use centre and scale factor

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{new point} = (1,2) + \frac{1}{2} (\text{old point} - (1,2))$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q8

Use

$$\text{new point} = (1,2) + \frac{1}{2} (\text{old point} - (1,2))$$

The vertices

$$(7,10), (15,10), (15,4), (11,4)$$

Finish: draw the enlarged shape with vertices $(4,6), (8,6), (8,3), (6,3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Write the vector translation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 3

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q1

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale factor is

3

Finish: (a) enlargement, scale factor 3, centre (3,0); (b) vertices (1,4), (2,4), (1,6).

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

Answer ALL TWENTY FIVE questions. Write your answers in the spaces provided.
You must write down all the stages in your working. 1 O x y 10 8 6 4 2 A B 2 4 6 8 10 12
14 (a) Describe fully the single transformation that maps triangle A onto triangle B (3)
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ Label your triangle C (1)

YOUR SOLUTION

04 Describe a combined movement

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(x, y) \mapsto (-6 - x, 4 - y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q5

(a) For a rotation about

$$(x, y) \mapsto (-6 - x, 4 - y)$$

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

Finish: (a) vertices $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$; (b) vertices $(7, 3), (13, 3), (13, 6), (10, 6)$; (c) enlargement, scale factor 2, centre $(-3, 3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 Here is a shape P drawn on a grid of squares. x y 2 4 6 8 2 4 6 8 10 12 14 O -2 -4 -6
-8 -2 -4 -6 -8 -10 P 10 (a) On the grid, rotate shape P 180 degrees about the point (-3, 2)
Label the new shape Q. (2) (b) On the grid, translate shape P by the vector 10 8 Label
the new shape R. (1) Here are triangle T and triangle U drawn on a grid of squares.
x y 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 10 11 12 O -1 -2 -3 -4 T U (c) Describe fully the single
transformation that maps triangle T onto triangle U. (3)

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(x, y) \mapsto (-6 - x, 4 - y)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q5

(a) For a rotation about

$$(x, y) \mapsto (-6 - x, 4 - y)$$

The image has vertices

$$(-3, 9), (-9, 9), (-9, 6), (-6, 6)$$

Finish: (a) vertices $(-3, 9), (-9, 9), (-9, 6), (-6, 6)$; (b) vertices $(7, 3), (13, 3), (13, 6), (10, 6)$; (c) enlargement, scale factor 2, centre $(-3, 3)$.

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

5 Here is a shape P drawn on a grid of squares. x y 2 4 6 8 2 4 6 8 10 12 14 O -2 -4 -6
-8 -2 -4 -6 -8 -10 P 10 (a) On the grid, rotate shape P 180 degrees about the point (-3, 2)
Label the new shape Q. (2) (b) On the grid, translate shape P by the vector 10 8 Label
the new shape R. (1) Here are triangle T and triangle U drawn on a grid of squares.
x y 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 10 11 12 O -1 -2 -3 -4 T U (c) Describe fully the single
transformation that maps triangle T onto triangle U. (3)

YOUR SOLUTION

06 Use Transformations tag correct

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 3

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q1

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale factor is

3

Finish: (a) enlargement, scale factor 3, centre (3,0); (b) vertices (1,4), (2,4), (1,6).

FORMULA BANK

Translation $\begin{pmatrix} a \\ b \end{pmatrix}$, reflection mirror line, rotation centre+angle, enlargement centre+scale factor.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

Answer ALL TWENTY FIVE questions. Write your answers in the spaces provided.
You must write down all the stages in your working. 1 O x y 10 8 6 4 2 A B 2 4 6 8 10 12
14 (a) Describe fully the single transformation that maps triangle A onto triangle B (3)
(b) On the grid above, translate triangle A by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$ Label your triangle C (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Identify the transformation
"work out", "show", "hence", a familiar exam phrase	Use centre and scale factor
"work out", "show", "hence", a familiar exam phrase	Write the vector translation
"work out", "show", "hence", a familiar exam phrase	Describe a combined movement
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use Transformations tag correct

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Statistics Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

STATISTICS TOOLKIT · CH6 · L21

ENG. ESLAM AHMED



The Map

This lesson collapses Statistics Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Choose the average

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use frequency table totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Estimate the mean

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Interpret spread

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 58 website questions, 58 checked solutions, 4 local PDFs read.

01 Choose the average

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w < 3$	12
$3 < w < 4$	16
$4 < w < 5$	9
$5 < w < 6$	2
$6 < w < 7$	1

(a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) (c) One of the 40 babies is going to be chosen at random. Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

02 Use frequency table totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w < 3$	12
$3 < w < 4$	16
$4 < w < 5$	9
$5 < w < 6$	2
$6 < w < 7$	1

(a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) (c) One of the 40 babies is going to be chosen at random. Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

03 Estimate the mean

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w < 3$	12
$3 < w < 4$	16
$4 < w < 5$	9
$5 < w < 6$	2
$6 < w < 7$	1

(a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) (c) One of the 40 babies is going to be chosen at random. Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

04 Interpret spread

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

5, 7, 11, 12, 13, 14, 15, 16, 17, 18, 18

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q14

Put the marks in order

5, 7, 11, 12, 13, 14, 15, 16, 17, 18, 18

Finish: 6.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

14 Max kept a record of the marks he scored in each of the 11 spelling tests he took one term. Here are his marks. 18 5 7 12 11 18 15 16 17 13 14 Find the interquartile range of the marks.

YOUR SOLUTION

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q2

Calculate statistic

$$a = 7$$

The median of four numbers

$$\frac{a + b}{2}$$

Finish: $a = 7, b = 10, c = 12$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

2 Given that $a < b < c$ the four whole numbers a, a, b and c have a mode of 7 a median of 8.5 a mean of 9 Work out the value of a , the value of b and the value of c . $a = b = c =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Choose the average
"work out", "show", "hence", a familiar exam phrase	Use frequency table totals
"work out", "show", "hence", a familiar exam phrase	Estimate the mean
"work out", "show", "hence", a familiar exam phrase	Interpret spread
"work out", "show", "hence", a familiar exam phrase	Calculate statistic

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Histograms

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

HISTOGRAMS · CH6 · L18

ENG. ESLAM AHMED



The Map

This lesson collapses Histograms into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use frequency density

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find missing frequency

TRIGGER *"work out", "find", a missing value*

03 Read area under a bar

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Build the histogram scale

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use histogram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use frequency table

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 33 website questions, 33 checked solutions, 3 local PDFs read.

01 Use frequency density

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5(40) + 10(20) = 400$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q19

Use the relative histogram heights

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate area

$$5(40) + 10(20) = 400$$

Finish: 370 students.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

02 Find missing frequency

TRIGGER *"work out", "find", a missing value*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5(40) + 10(20) = 400$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q19

Use the relative histogram heights

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate area

$$5(40) + 10(20) = 400$$

Finish: 370 students.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Read area under a bar

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1, 4, 9, 6, 1

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q18

Use histogram

1, 4, 9, 6, 1

Use histogram

28, 112, 126, 84, 42

Finish: 4.36 hours.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

04 Build the histogram scale

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM1H Q20

Frequency density

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

Finish: draw bars with densities 2.4, 3.4, 3, 1.4.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

05 Use histogram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 1, 4, 9, 6, 1

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q18

Use histogram

1, 4, 9, 6, 1

Use histogram

28, 112, 126, 84, 42

Finish: 4.36 hours.

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

06 Use frequency table

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q21

Use histogram

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

Use frequency table

$$\text{frequency density} = \frac{16}{0.5} = 32$$

Finish: 102 watermelons

FORMULA BANK

Frequency density = $\frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use frequency density
"work out", "find", a missing value	Find missing frequency
"work out", "show", "hence", a familiar exam phrase	Read area under a bar
"work out", "show", "hence", a familiar exam phrase	Build the histogram scale
"work out", "show", "hence", a familiar exam phrase	Use histogram
"work out", "show", "hence", a familiar exam phrase	Use frequency table

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Cumulative Frequency Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CUMULATIVE FREQUENCY DIAGRAMS · CH6 · L17

ENG. ESLAM AHMED



The Map

This lesson collapses Cumulative Frequency Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Build cumulative totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Read median and quartiles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Find IQR

TRIGGER *"work out", "find", a missing value*

04 Use percentages from the curve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use cumulative frequency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 24 website questions, 24 checked solutions, 3 local PDFs read.

01 Build cumulative totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Q_1 is at cumulative frequency 20

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q12

There are people

This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

Q_1 is at cumulative frequency 20

Finish: IQR about 18 minutes, and about 70 men.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 A total of 80 men and women took part in a race. The cumulative frequency graph gives information about the times, in minutes, they took for the race.

80 70 60 50 40 30 20 10 0 20 30 40 50 60 70 Time (minutes) Cumulative frequency

(a) Use the graph to find an estimate for the interquartile range. minutes (2) 60% of the men took 50 minutes or less for the race. No women took 50 minutes or less for the race. (b) Work out an estimate for the number of men who took part in the race. (3)

YOUR SOLUTION

02 Read median and quartiles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

median $\approx$ 22.5 minutes

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q12

There are people

This is a reasoning step: write one clear sentence, then continue the calculation.

The median is at cumulative

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: median about 22.5 minutes, IQR about 12 minutes. Hospital A is quicker and more consistent.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Find IQR

TRIGGER *"work out", "find", a missing value*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

median ≈ 146 g

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q11

Use cumulative frequency

This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

median ≈ 146 g

Finish: median about 146 g, and $d \approx 150$.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.

Weight (grams)	Cumulative frequency
0	0
10	10
20	20
30	30
40	40
50	50
60	60
70	70
80	80
90	90

(a) Find an estimate for the median of the weights of these bags of sweets.

(2) Roberto sells the bags of sweets to raise money for charity. Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag. The total amount of money he receives by selling all the large bags is 93.75 ...

YOUR SOLUTION

04 Use percentages from the curve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM2HR Q11

Required

This is a reasoning step: write one clear sentence, then continue the calculation.

Median at the th value

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: Alison is more consistent — her IQR (15 km) is smaller than Tomas's (28 km), meaning less variability in daily distances.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

11 The cumulative frequency graph shows information about the distance, in kilometres, Ehrick drives each day for 60 days.

distance (kilometres)	Cumulative frequency
0	0
10	10
20	20
30	30
40	40
50	50
60	50
70	40
80	30

(a) Use the graph to find an estimate for the median of the distances. kilometres (1)

(b) Use the graph to find an estimate for the interquartile range of the distances. kilometres (2)

(c) Work out the percentage of days that Ehrick drives for more than 75 kilometres. Give your answer correct to one decimal ...

YOUR SOLUTION

05 Use cumulative frequency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Q_1 is at cumulative frequency 20

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q12

There are people

This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

Q_1 is at cumulative frequency 20

Finish: IQR about 18 minutes, and about 70 men.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

12 A total of 80 men and women took part in a race. The cumulative frequency graph gives information about the times, in minutes, they took for the race.

Time (minutes)	Cumulative frequency
0	0
10	20
20	30
30	40
40	50
50	60
60	70
70	80

(a) Use the graph to find an estimate for the interquartile range.

(2) 60% of the men took 50 minutes or less for the race. No women took 50 minutes or less for the race.

(b) Work out an estimate for the number of men who took part in the race.

(3)

06 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

7, 33, 57, 71, 78, 80

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q12

Complete the cumulative frequencies

7, 33, 57, 71, 78, 80

Plot the points

$(0, 0), (10, 7), (20, 33), (30, 57), (40, 71), (50, 78), (60, 80)$

Finish: cumulative frequencies 7, 33, 57, 71, 78, 80, median about 23 minutes, probability about 0.095.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket. Time taken (t minutes) Frequency

$0 < t < 10$	7
$10 < t < 20$	26
$20 < t < 30$	24
$30 < t < 40$	14
$40 < t < 50$	7
$50 < t < 60$	2

(a) Complete the cumulative frequency table. Time taken (t minutes) Cumulative frequency

$0 < t < 10$	
$0 < t < 20$	
$0 < t < 30$	
$0 < t < 40$	
$0 < t < 50$	
$0 < t < 60$	

(1) (b) On the grid opposite, draw a cumulative frequency graph for your table.

0	10	20	30	40	50	60
Cumulative frequency						

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Build cumulative totals
"work out", "show", "hence", a familiar exam phrase	Read median and quartiles
"work out", "find", a missing value	Find IQR
"work out", "show", "hence", a familiar exam phrase	Use percentages from the curve
"work out", "show", "hence", a familiar exam phrase	Use cumulative frequency
"work out", "show", "hence", a familiar exam phrase	Convert standard form
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Probability Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PROBABILITY TOOLKIT · CH6 · L20



ENG. ESLAM AHMED

The Map

This lesson collapses Probability Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 45 website questions, 45 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{x}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q24

There are counters. Initially 5

This is a reasoning step: write one clear sentence, then continue the calculation.

List probability cases

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $x = 8$

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$r + 0.24 + 2r + 0.31 = 1$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q6

Let the probability of red

$$r + 0.24 + 2r + 0.31 = 1$$

Expected number of red counters

$$180 \times 0.15 = 27$$

Finish: 27.

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| 6

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q21

There are beads altogether

This is a reasoning step: write one clear sentence, then continue the calculation.

Red beads

| 6
Finish: 18

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{no lemon}) = \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q18

There are sweets and are

$$P(\text{no lemon}) = \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14}$$

Finish: $\frac{33}{112}$.

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0.65 \times 300 = 195$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q2

Calculate probability

$$0.65 \times 300 = 195$$

Finish: 195.

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Probability Diagrams - Venn & Tree Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PROBABILITY DIAGRAMMS - VENN & TREE DIAGRAMMS · CH6 · L19

ENG. ESLAM AHMED



The Map

This lesson collapses Probability Diagrams - Venn & Tree Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q16

Calculate probability

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

Calculate probability

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Finish: $\frac{7}{15}$, and 25.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q16

Calculate probability

$$P(A \text{ odd}) = \frac{7}{10}, \quad P(A \text{ even}) = \frac{3}{10}$$

Calculate probability

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Finish: $\frac{7}{15}$, and 25.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0.6 \times 0.9 = 0.54$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q14

Winning both frames

$$0.6 \times 0.9 = 0.54$$

Winning exactly one frame

$$0.6 \times 0.1 + 0.4 \times 0.25$$

Finish: 0.54, 0.16.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q14

Calculate probability

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

Finish: $\frac{21}{190}$.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q14

Calculate probability

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

Finish: $\frac{21}{190}$.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Combined & Conditional Probability

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMBINED & CONDITIONAL PROBABILITY · CH6 · L16

ENG. ESLAM AHMED



The Map

This lesson collapses Combined & Conditional Probability into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

06 Use the complement

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 11 website questions, 11 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q20

There are 3 red counters

This is a reasoning step: write one clear sentence, then continue the calculation.

At least one red counter

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{odd}) = \frac{1}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q14

The complement of at least

$$P(\text{odd}) = \frac{1}{2}$$

Finish: $\frac{3}{4}$.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q20

There are 3 red counters

This is a reasoning step: write one clear sentence, then continue the calculation.

At least one red counter

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{win tennis}) = 0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q16

Calculate probability

$$P(\text{win tennis}) = 0.6$$

Calculate probability

$$0.42 \div 0.6 = 0.7$$

Finish: 0.12.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{win tennis}) = 0.6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q16

Calculate probability

$$P(\text{win tennis}) = 0.6$$

Calculate probability

$$0.42 \div 0.6 = 0.7$$

Finish: 0.12.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

06 Use the complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q20

There are 3 red counters

This is a reasoning step: write one clear sentence, then continue the calculation.

At least one red counter

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability
"probability", "random", "given that"	Use the complement
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.